

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

CS407

[Routing and switching]

[Reading material]

WEEK [1 TO 6]

TOPIC [1 TO 66]

Made by Mah jabeen

[mahjabeen97869@gmail.com]

1. TOPIC

The focus of this course is education and to help you develop a real understanding of networking, not merely memorizing facts and commands.

We have divided this course into three parts:

1. Network Fundamentals

In this part, you will learn both the practical and conceptual skills that build the foundation for understanding basic networking.

2. Routing Protocols and Concepts

The focus of this part is on learning the architecture, components, and operations of routers, and explains the principles of routing and routing protocols.

3. Switching Techniques

The focus of this part is on learning the architecture, components, and operations of a converged switched network.

By the end of this course, you will be able to build simple LANs, perform basic configurations for routers and switches, and implement IP addressing schemes.

2. Topic

Technology Then and Now:

Communication is almost as important to us as our reliance on air, water, food, and shelter. Communicate with friends, family, and peers using email, instant messaging, or Internet phone calls.

Internet of things:

All devices near us can be accessed with the help of internet of things by assigning of the IP address known as **IPv4** address.

Network Support the way of learning

Support the creation of virtual classrooms

Provide on-demand video

Enable collaborative learning spaces

Enable mobile learning

Access to high quality instruction is no longer restricted to students living in proximity to where that instruction is being delivered.

Network Support the way of Communication

Instant Messaging (IM) / Texting

IM and texting both enable instant real-time communication between two or more people. IM applications can offer additional features such as voice and video communication.

Social Media

Social media consists of interactive websites where people and communities create and share user-generated content with friends, family, peers, and the world.

Collaboration Tools - Collaboration tools give people the opportunity to work together on shared documents.

Weblogs

Weblogs are web pages that are easy to update and edit. Unlike commercial websites, which are created by professional communications experts, blogs give anyone a means to communicate their thoughts to a global audience.

Podcasting - Podcasting is an audio-based medium that originally enabled people to record audio and convert it for use.

Network Support the way of Business/Gaming

Online Gaming

With the help of networks, you can play games from anywhere in the world.

Online Shopping

With the help of networks, you can buy online from anywhere in the world. Like Daraz, Amazon, Etsy.

Online Entertainment

With the help of networks, you can now stream from any entertainment website like AmazonPrime, Netflix etc.

3. Topic

Providing Resources in a Network:

1-Network Sizes:

Small Home Network

Network in the home premises i.e. Network spread in a home via ISP.
3-5 computers connected.

Small Office Network

Network in a small scale office premises i.e Network spread in an office via some routers etc.
15-25 computers connected.

Medium to Large

Network in a university or a corporate company that possess 100-300 computers via routers and switeches.

World Wide Networks

The mighty INTERNET.

2-Network Components:

Hosts Server & Client:

All computers connected to a network that participate directly in network communication are classified as **hosts or end devices**. Hosts can send and receive messages on the network. In modern networks, end devices can act as a **client, a server**, or both. The software installed on the computer determines which role the computer plays.

Servers are hosts that have software installed that enable them to provide information, like email or web pages, to other hosts on the network. Each service requires separate server software. For example, a host requires web server software in order to provide web services to the network.

Peer to Peer:

Client and server software usually runs on separate computers, but it is also possible for one computer to carry out both roles at the same time. In small businesses and homes, many computers function as the servers and clients on the network. This type of network is called a peer-to-peer network.

Categories of Network Components:

The network infrastructure contains three categories of network components:

- Devices
- Media
- Services

Devices and media are the physical elements, or hardware, of the network. Hardware is often the visible components of the network platform such as a laptop, PC, switch, router, wireless access point, or the cabling used to connect the devices.

Devices are of two types:

- **End Devices**

Some examples of end devices are:

Computers (work stations, laptops, file servers, web servers), Network printers, VoIP phones, Security cameras

- **Intermediate Devices**

Intermediary devices interconnect end devices. These devices provide connectivity and work behind the scenes to ensure that data flows across the network, as shown in the animation. Intermediary devices connect the individual hosts to the network and can connect multiple individual networks to form an internetwork.

Examples of intermediary network devices are:

Network Access (switches and wireless access points)

Network Media is basically:

The medium provides the channel over which the message travels from source to destination. Modern networks primarily use three types of media to interconnect devices and to provide the pathway over which data can be transmitted. Media are of three types that are commonly used:

- Metallic wires within cables
- Glass or plastic fibers (fiber optic cable)
- Wireless transmission

Topology diagrams:

These diagrams are mandatory for anyone working with a network. Its objective is to provide a visual map of how the network is connected. There are two types of topology diagrams including:

- **Physical topology diagrams** - Identify the physical location of intermediary devices, configured ports, and cable installation.
- **Logical topology diagrams** - Identify devices, ports, and IP addressing scheme.

4. Topic

Types of networks:

Local Area Network (LAN) - A network infrastructure that provides access to users and end devices in a small geographical area. LANs interconnect end devices in a limited area such as a home, school, office building, or campus. LANs provide high speed bandwidth to internal end devices and intermediary devices.

Wide Area Network (WAN) - A network infrastructure that provides access to other networks over a wide geographical area. WANs interconnect LANs over wide geographical areas such as between cities, states, provinces, countries, or continents. WANs are usually administered by multiple service providers. WANs typically provide slower speed links between LANs.

Metropolitan Area Network (MAN) - A network infrastructure that spans a physical area larger than a LAN but smaller than a WAN (e.g., a city). MANs are typically operated by a single entity such as a large organization.

Wireless LAN (WLAN) - Similar to a LAN but wirelessly interconnects users and end points in a small geographical area.

Storage Area Network (SAN) - A network infrastructure designed to support file servers and provide data storage, retrieval, and replication. It involves high-end servers and Fiber Channel technology.

The Internet - Internet is a network of networks.

Internet Access Technologies:

We can access the internet via the following technologies:

- Broadband cable
- Broadband digital subscriber line (DSL)
- Wireless WANs
- Mobile services (3G, 4G)
- Dial-Up Modem
- Satellite Connection

5. Topic

Establishing the rules:

Communication begins with a message, or information, that must be sent from a source to a destination

Protocol: Set of rules that govern communications

Protocol suite: A group of inter-related protocols i.e. **TCP/IP**

Message Encoding

One of the first steps to sending a message is encoding it. Encoding is the process of converting information into another, acceptable form, for transmission.

Message Formatting

When a message is sent from source to destination, it must use a specific format or structure. Message formats depend on the type of message and the channel that is used to deliver the message.

Message Encapsulation

The process of placing one message format (the letter) inside another message format is called encapsulation.

Message Size

When people communicate with each other, the messages that they send are usually broken into smaller parts or sentences. These sentences are limited in size to what the receiving person can process at one time.

Message Timing

Another factor that affects how well a message is received and understood is timing. People use timing to determine when to speak, how fast or slow to talk, and how long to wait for a response. These are the rules of engagement.

Access Method

Access method determines when someone is able to send a message. These timing rules are based on the environment. For example, you may be able to speak whenever you have something to say.

Flow Control

Timing also affects how much information can be sent and the speed that it can be delivered. If one person speaks too quickly, it is difficult for the other person to hear and understand the message.

Response Timeout

If a person asks a question and does not hear a response within an acceptable amount of time, the person assumes that no answer is coming and reacts accordingly.

Message Delivery Options:

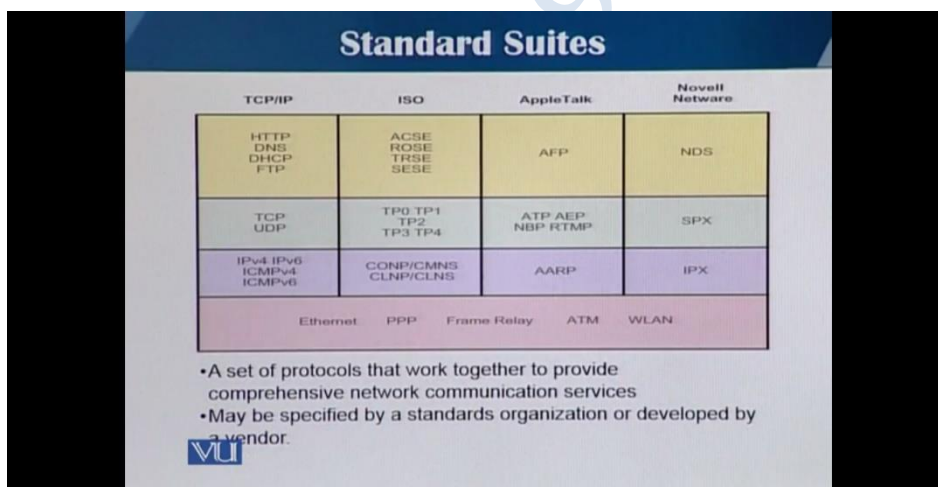
- **Unicast**
Sending data to the single machine.
- **Multicast**
Sending data to multiple machines in a network
- **Broadcast**

Sending data to all units in a network

6. Topic

TCP/IP Protocol Suite

A protocol suite is a set of protocols that work together to provide comprehensive network communication services. A protocol suite may be specified by a standards organization or developed by a vendor.



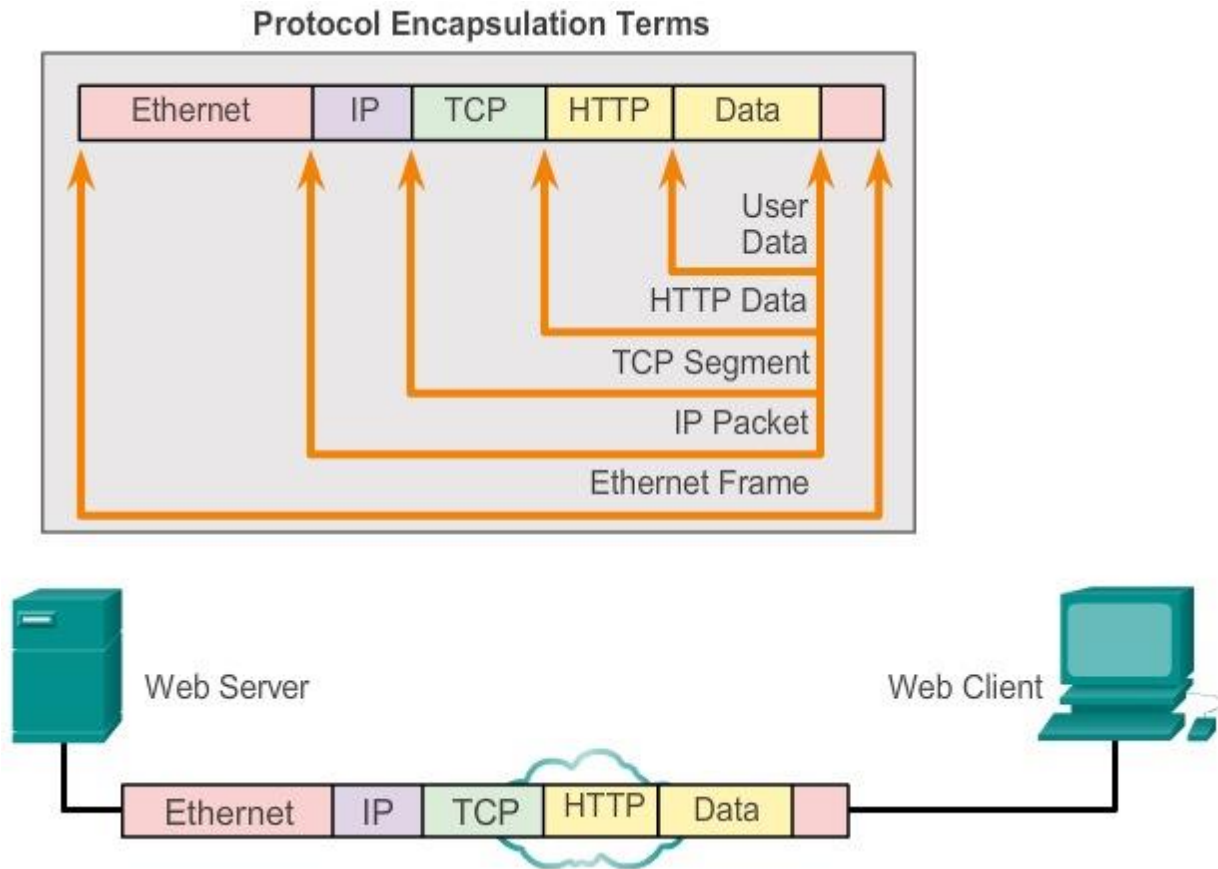
The protocols IP, HTTP, and DHCP are all part of the Internet protocol suite known as Transmission Control Protocol/IP (TCP/IP). The TCP/IP protocol suite is an open standard, meaning these protocols are freely available to the public, and any vendor is able to implement these protocols on their hardware or in their software.

Why it is called TCP/IP Protocol:

It is commonly known as TCP/IP because the first two networking protocols defined for this standard were TCP and IP.

The open standards-based TCP/IP has replaced other vendor proprietary protocol suites, such as AppleTalk and Novella Internetwork Packet Exchange/Sequenced Packet Exchange.

Real time Example of accessing a web page from a web server:



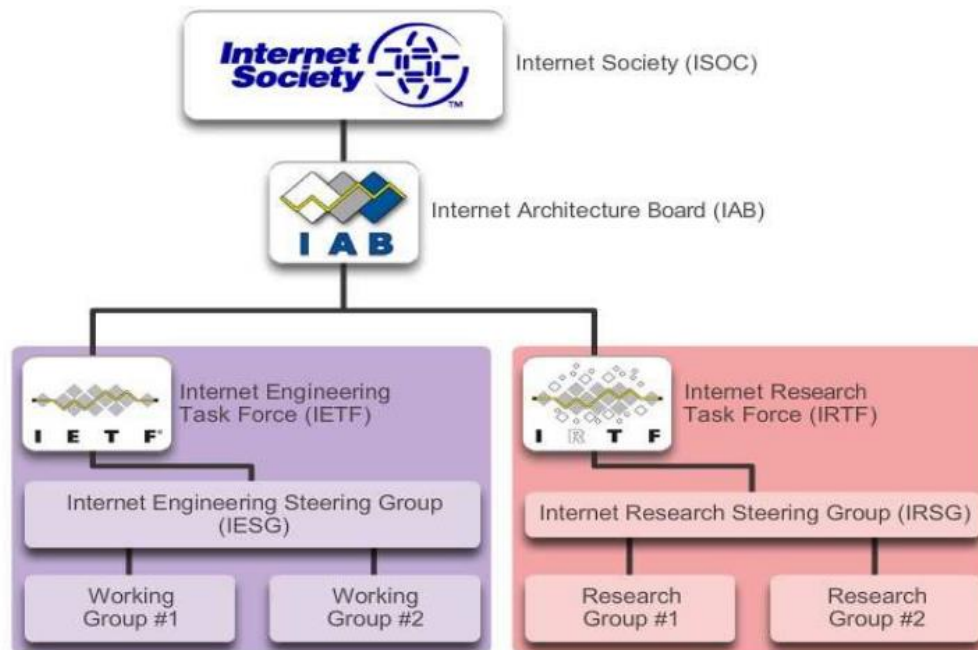
- The web server Hypertext Markup Language (HTML) page is the data to be sent.
- The application protocol HTTP header is added to the front of the HTML data. The header contains various information, including the HTTP version the server is using and a status code indicating it has information for the web client.
- The HTTP application layer protocol delivers the HTML-formatted web page data to the transport layer. The TCP transport layer protocol is used to manage the individual conversation between the web server and web client.
- Next, the IP information is added to the front of the TCP information. IP assigns the appropriate source and destination IP addresses. This information is known as an IP packet.
- The Ethernet protocol adds information to both ends of the IP packet, known as a data link frame. This frame is delivered to the nearest router along the path towards the web client. This router removes the Ethernet information, analyzes the IP packet, determines the best path for the packet, inserts the packet into a new frame, and sends it to the next neighboring router towards the destination. Each router removes and adds new data link information before forwarding the packet.
- This data is now transported through the internetwork, which consists of media and intermediary devices.
- The client receives the data link frames that contain the data and each protocol header is processed and then removed in the opposite order that it was added. The Ethernet information is processed and removed, followed by the IP protocol information, then the TCP information, and finally the HTTP information.
- The web page information is then passed on to the client's web browser software.

7. Topic

Standard Organizations

Open standards encourage competition and innovation.

Guarantee that no single company's product can monopolize the market, or have an unfair advantage over its competition.



Standards organizations include:

- The Internet Society (ISOC)
- The Internet Architecture Board (IAB)
- The Internet Engineering Task Force (IETF)
- The Institute of Electrical and Electronics Engineers (IEEE)
- The International Organization for Standardization (ISO)

The Internet Society (ISOC)

- ISOC facilitates the open development of standards and protocols for the technical infrastructure of the Internet, including the oversight of the Internet Architecture Board (IAB).

The Internet Architecture Board (IAB)

- Responsible for overall management and development of Internet standards.
- Oversight of the architecture for protocols and procedures used by the Internet.
- 13 members, including the chair of the Internet Engineering Task Force (IETF).
- IAB members serve as individuals and not representatives of any company, agency, or other organization.

Internet Engineering Task Force (IETF)

- Mission is to develop, update, and maintain Internet and TCP/IP technologies.
- One of the key responsibilities is to produce Request for Comments (RFC) documents.
- Memorandum describing protocols, processes, and technologies for the Internet.
- The IETF consists of working groups (WGs), the primary mechanism for developing IETF specifications and guidelines.

Internet Research Task Force (IRTF)

- Focused on long-term research related to Internet and TCP/IP
- IETF focuses on shorter-term issues of creating standards
- IRTF consists of research groups for long-term development efforts. Including: Anti-Spam Research Group (ASRG), Crypto Forum Research Group (CFRG), Peer-to-Peer Research Group (P2PRG), and Router Research Group (RRG).

Institute of Electrical and Electronics Engineers - IEEE

- A professional organization for those in the electrical engineering and electronics fields who are dedicated to advancing technological innovation and creating standards.
- IEEE 802.3 and IEEE 802.11 standards are significant IEEE standards in computer networking.

8. TOPIC

Reference Models

There are 2 types of layered reference models:

- OSI Models
- TCP/IP Models

OSI model consist of 7 layers (Application, Presentation, Session, Transport, Network, Datalink, Physical)

TCP/IP model consist of 4 layers (Application, Transport, Network, Network Access)

Benefits of Layered Model

Management is easy when considering layers.

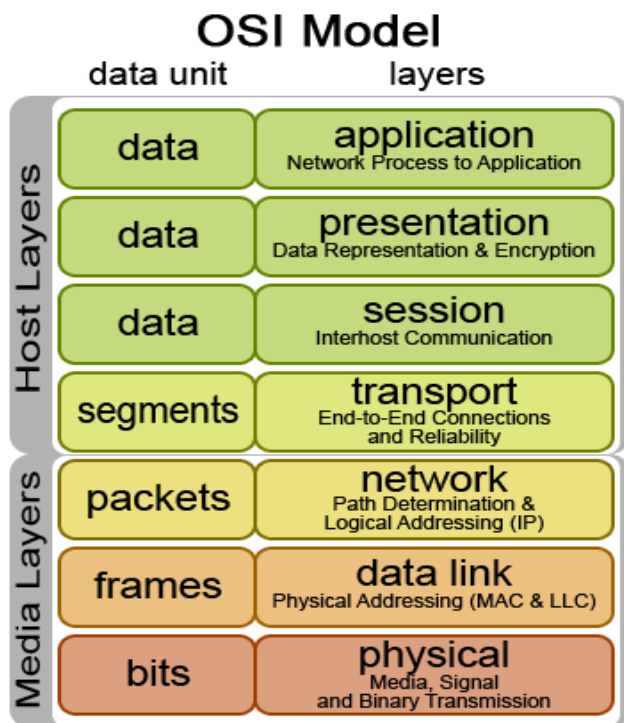
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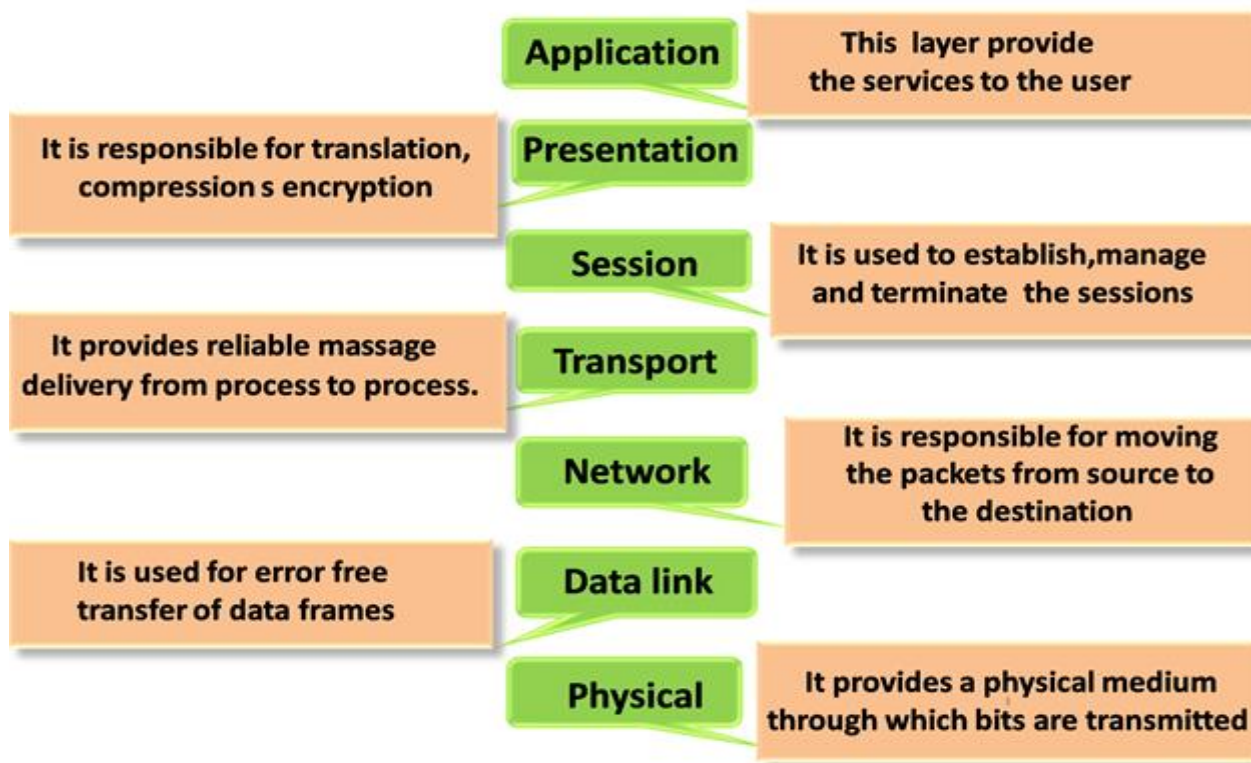
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Made by Mah jabeen

[mahjabeen97869@gmail.com]

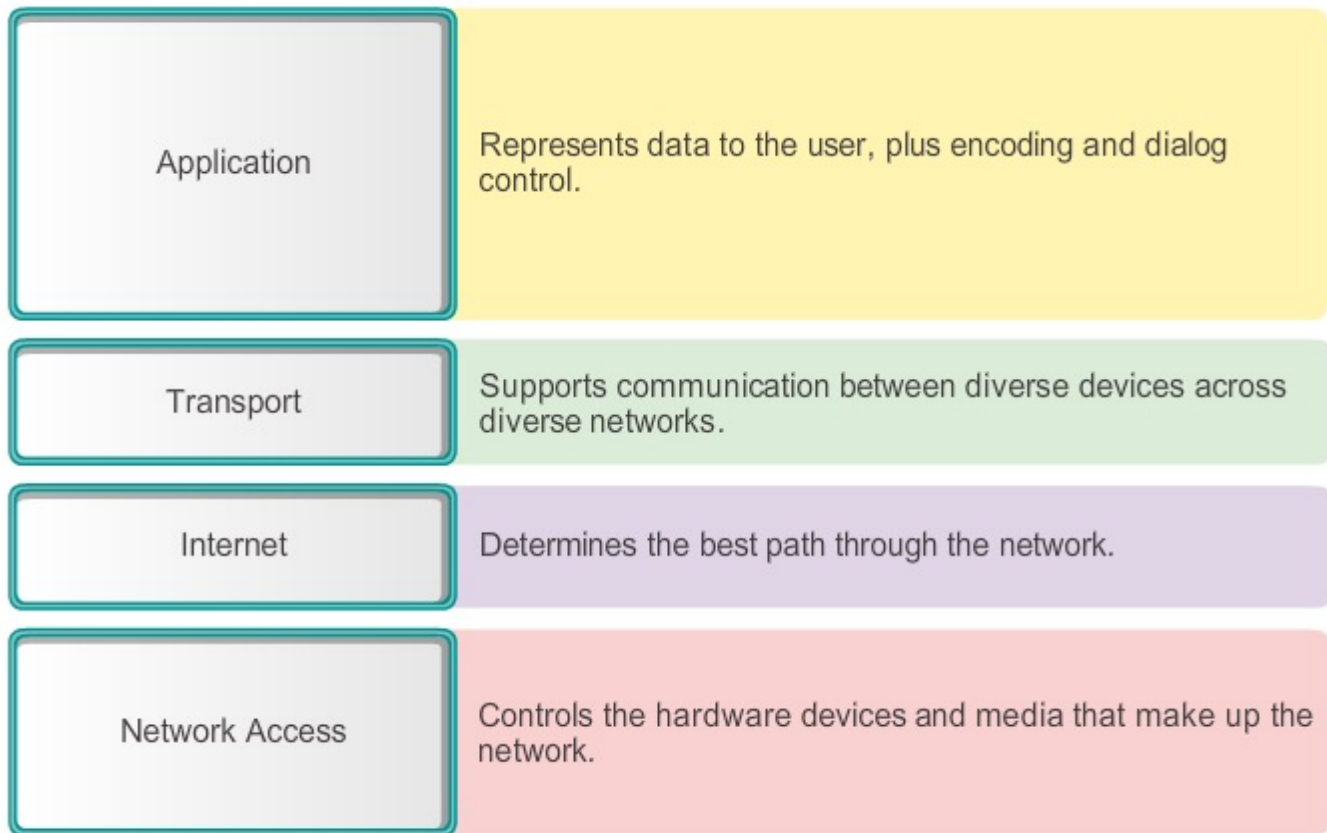


Here is the functioning of OSI layers explained in the image below:



Here is the functioning of TCP/IP layers explained in the image below:

TCP/IP Model



9.TOPIC

Cisco IOS

Infrastructure network devices use a network operating system. The network operating system used on Cisco devices is called the Cisco Internetwork Operating System (IOS). Cisco IOS is a generic term for the collection of network operating systems used on Cisco networking devices.

Layers of OS

O.S is divided into 3 layers:

- **Shell**
It is a user interface that allows a user to make specific task from the computer.
- **Kernel**
It communicates between hardware and software of a computer and manages how hardware resources are used.
- **Hardware**
The physical part of the computer.

CISCO IOS Functions

Cisco IOS routers and switches perform functions that network professionals depend upon to make their networks operate as expected. Major functions performed or enabled by Cisco routers and switches include:

- Providing network security

- IP addressing of virtual and physical interfaces
- Enabling interface-specific configurations to optimize connectivity of the respective media
- Routing
- Enabling quality of service (QoS) technologies
- Supporting network management technologies

Example of a network device:

All of the network equipment devices requires operating system to work and considering the example of a home router. Home routers are actually four devices in one:

Router - Forwards data packets to and receives data packets from the Internet

Switch - Connects end devices using network cables

Wireless access point - Consists of a radio transmitter capable of connecting end devices wirelessly

Firewall appliance - Secures outgoing traffic and restricts incoming traffic

10. Topic

Control Access Method

There are several ways to access the CLI environment. The most common methods are:

- Console
- Telnet or SSH
- AUX port

Console Port

The console port is a management port that provides out-of-band access to Cisco device. Out-of-band access refers to access via a dedicated management channel that is used for device maintenance purposes only. The advantage of using a console port is that the device is accessible even if no networking services have been configured, such as when performing an initial configuration of the networking device. When performing an initial configuration, a computer running terminal emulation software is connected to the console port of the device using a special cable. Configuration commands for setting up the switch or router can be entered on the connected computer.

Telnet

Telnet is a method for remotely establishing a CLI session of a device, through a virtual interface, over a network. Unlike the console connection, Telnet sessions require active networking services on the device. The network device must have at least one active interface configured with an Internet address, such as an IPv4 address. Cisco IOS devices include a Telnet server process that allows users to enter configuration commands from a Telnet client. In addition to supporting the Telnet server process, the Cisco IOS device also contains a Telnet client. This allows a network administrator to telnet from the Cisco device CLI to any other device that supports a Telnet server process.

SSH

The Secure Shell (SSH) protocol provides a remote login similar to Telnet, except that it uses more secure network services. SSH provides stronger password authentication than Telnet and uses encryption when transporting session data. This keeps the user ID, password, and the details of the management session private. As a best practice, use SSH instead of Telnet whenever possible.

AUX

An older way to establish a CLI session remotely is via a telephone dialup connection using a modem connected to the auxiliary (AUX) port of a router, which is highlighted in the figure. Similar to the console connection, the AUX method is also an out-of-band connection and does not require any networking services to be configured or available on the device.

In the event that network services have failed, it may be possible for a remote administrator to access the switch or router over a telephone line.

11. Topic

IOS modes of Operations

After a network technician is connected to a device, it is possible to configure it. The network technician must navigate through various modes of the IOS. The Cisco IOS modes are quite similar for switches and routers. The CLI uses a hierarchical structure for the modes.

In hierarchical order from most basic to most specialized, the major modes are:

- **User executive (User EXEC) mode**
- **Privileged executive (Privileged EXEC) mode**
- **Global configuration mode**

The two primary modes of operation are user EXEC mode and privileged EXEC mode. As a security feature, the Cisco IOS software separates the EXEC sessions into two levels of access. As shown in the figure, the privileged EXEC mode has a higher level of authority in what it allows the user to do with the device.

User EXEC Mode Description:

- The user EXEC mode has limited capabilities but is useful for some basic operations. The user EXEC mode is at the most basic level of the modal hierarchical structure. This mode is the first mode encountered upon entrance into the CLI of an IOS device.
- The user EXEC mode allows only a limited number of basic monitoring commands. This is often referred to as view-only mode. The user EXEC level does not allow the execution of any commands that might change the configuration of the device.
- By default, there is no authentication required to access the user EXEC mode from the console. However, it is a good practice to ensure that authentication is configured during the initial configuration.
- The user EXEC mode is identified by the CLI prompt that ends with the > symbol. This is an example that shows the > symbol in the prompt: Switch>

Privileged EXEC Mode Description:

- The execution of configuration and management commands requires that the network administrator use the privileged EXEC mode or a more specific mode in the hierarchy. This means that a user must enter user EXEC mode first, and from there, access privileged EXEC mode.
- The privileged EXEC mode can be identified by the prompt ending with the # symbol. Switch#
- By default, privileged EXEC mode does not require authentication. It is a good practice to ensure that authentication is configured.
- Global configuration mode and all other more specific configuration modes can only be reached from the privileged EXEC mode. In a later section of this chapter, we will examine device configuration and some of the configuration modes.
- Global configuration mode and interface configuration modes can only be reached from the privileged EXEC mode.

Global Configuration Mode Description

- The primary configuration mode is called global configuration or global config. From global configuration mode, CLI configuration changes are made that affect the operation of the device as a whole. The global configuration mode is accessed before accessing specific configuration modes.

- The following CLI command is used to take the device from privileged EXEC mode to the global configuration mode and to allow entry of configuration commands from a terminal: Switch# **configure terminal**
- After the command is executed, the prompt changes to show that the switch is in global configuration mode. Switch(config)#

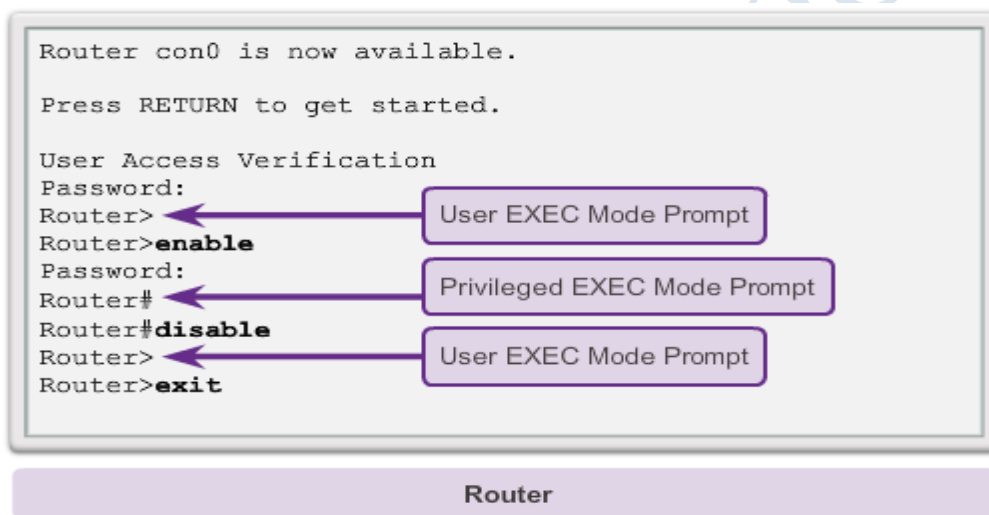
Specific Configuration Modes

- From the global configuration mode, the user can enter different sub-configuration modes. Each of these modes allows the configuration of a particular part or function of the IOS device. The list below shows a few of them:
- **Interface mode** - to configure one of the network interfaces (Fa0/0, S0/0/0)
- **Line mode** - to configure one of the physical or virtual lines (console, AUX, VTY)
- Figure 1 shows the prompts for some of these modes. To exit a specific configuration mode and return to global configuration mode, enter **exit** at a prompt. To leave configuration mode completely and return to privileged EXEC mode, enter **end** or use the key sequence **Ctrl-Z**.

Command Prompts

- When using the CLI, the mode is identified by the command-line prompt that is unique to that mode. By default, every prompt begins with the device name. Following the name, the remainder of the prompt indicates the mode. For example, the default prompt for the global configuration mode on a switch would be: Switch(config)#

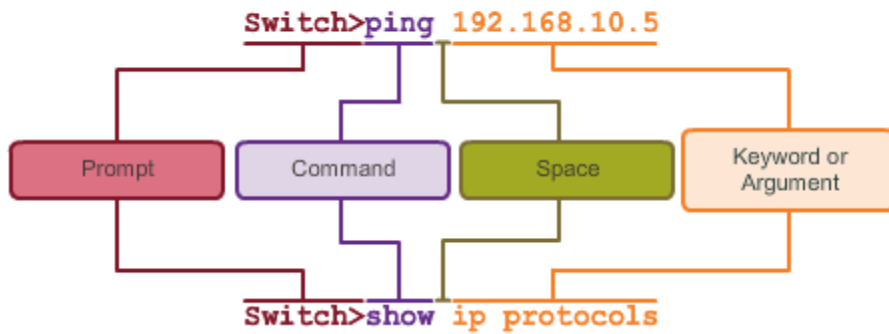
Navigating between IOS Modes



Week no 2

12. TOPIC

Basic IOS Command Structure



IOS Command Reference:

The Command Reference is a fundamental resource that network engineers use to check various characteristics of a given IOS command.

XX. For the **ping** command:

Switch> **ping** *IP-address*

Switch> **ping** 10.10.10.5

The command is **ping** and the user defined argument is the **10.10.10.5**

IOS Command Help:

The IOS has several forms of help available:

- Context-Sensitive Help
- Command Syntax Check
- Hot Keys and Shortcuts

Context-Sensitive Help

The context-sensitive help provides a list of commands and the arguments associated with those commands within the context of the current mode. To access context-sensitive help, enter a question mark, **?**, at any prompt. There is an immediate response without the need to use the Enter key.

Command Syntax Check

When a command is submitted by pressing the Enter key, the command line interpreter parses the command from left to right to determine what action is being requested. The IOS generally only provides negative feedback. If the interpreter understands the command, the requested action is executed and the CLI returns to the appropriate prompt. However, if the interpreter cannot understand the command being entered, it will provide feedback describing what is wrong with the command.

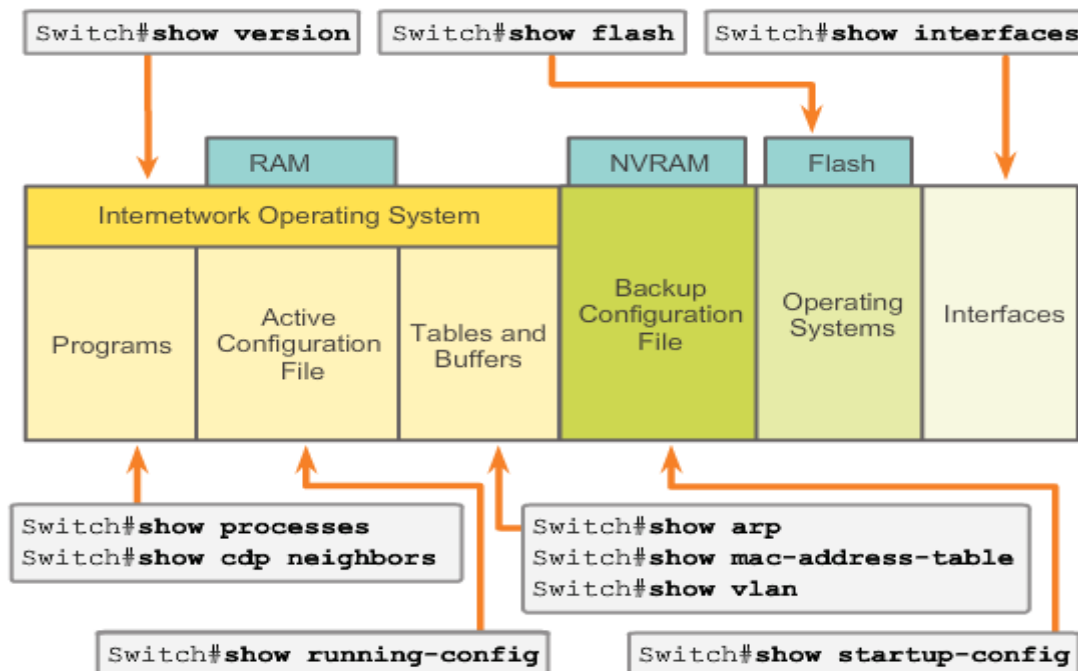
Hot Keys and Shortcuts

The IOS CLI provides hot keys and shortcuts that make configuring, monitoring, and troubleshooting easier.

- **Down Arrow** - Allows the user to scroll forward through former commands
- **Up Arrow** - Allows the user to scroll backward through former commands
- **Tab** - Completes the remainder of a partially typed command or keyword

- **Ctrl-A** - Moves to the beginning of the line
- **Ctrl-E** - Moves to the end of the line
- **Ctrl-R** - Redisplays a line
- **Ctrl-Z** - Exits the configuration mode and returns to user EXEC
- **Ctrl-C** - Exits the configuration mode or aborts the current command
- **Ctrl-Shift-6** - Allows the user to interrupt an IOS process such as ping or traceroute

IOS Examination Commands



13. TOPIC

Cisco IOS Switch

Cisco IOS switch is one of the simplest devices that can be configured on a network. This is because there are no configurations that are required prior to the device functioning. At its most basic, a switch can be plugged in with no configuration, but it will still switch data between connected devices.

A switch is also one of the fundamental devices used in the creation of a small network. By connecting two PCs to a switch, those PCs will instantly have connectivity with one another.

Device/Host Names Guidelines:

Hostnames allow devices to be identified by network administrators over a network or the Internet.

Some guidelines for naming conventions are that names should:

- Start with a letter
- Contain no spaces
- End with a letter or digit
- Use only letters, digits, and dashes
- Be less than 64 characters in length

Configuring Host

```
Configure the switch hostname to be 'Sw-Floor-1'.
Switch# configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)# hostname Sw-Floor-1
Sw-Floor-1(config)#
You successfully configured the switch hostname.
```

From the privileged EXEC mode, access the global configuration mode by entering the **configure terminal** command:

Switch# **configure terminal**

After the command is executed, the prompt will change to:

Switch(config)#

As shown in the above image, in the global configuration mode, enter the hostname:

Switch(config)# **hostname Sw-Floor-1**

After the command is executed, the prompt will change to:

Sw-Floor-1 (config)#

Notice that the hostname appears in the prompt. To exit global configuration mode, use the **exit** command.

Always make sure that your documentation is updated each time a device is added or modified. Identify devices in the documentation by their location, purpose, and address.

Securing Device Access

```
Sw-Floor-1>enable
Sw-Floor-1#
Sw-Floor-1#conf terminal
Sw-Floor-1(config)#enable secret class
Sw-Floor-1(config)#exit
Sw-Floor-1#
Sw-Floor-1#disable
Sw-Floor-1>enable
Password:
Sw-Floor-1#
```

- use the **enable secret** command, not the older **enable password** command
- **enable secret** provides greater security because the password is encrypted

```

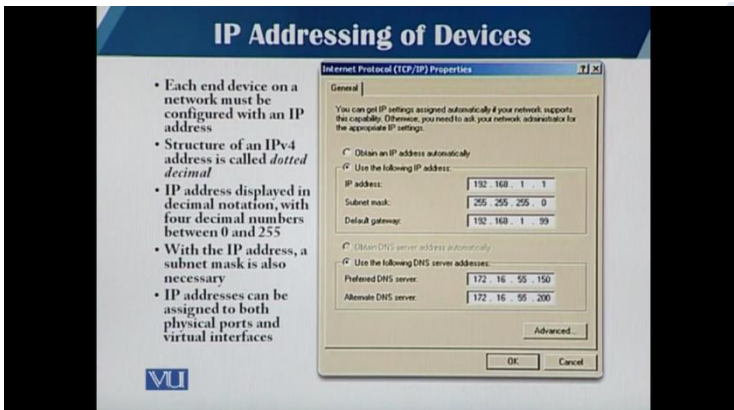
Sw-Floor-1 (config) #line console 0
Sw-Floor-1 (config-line) #password cisco
Sw-Floor-1 (config-line) #login
Sw-Floor-1 (config-line) #exit
Sw-Floor-1 (config) #
Sw-Floor-1 (config) #line vty 0 15
Sw-Floor-1 (config-line) #password cisco
Sw-Floor-1 (config-line) #login
Sw-Floor-1 (config-line) #

```

- **Console port** must be secured because
 - Reduces the chance of unauthorized personnel physically plugging a cable into the device and gaining device access
- **VTY lines** allow access to a Cisco device via Telnet

14. TOPIC

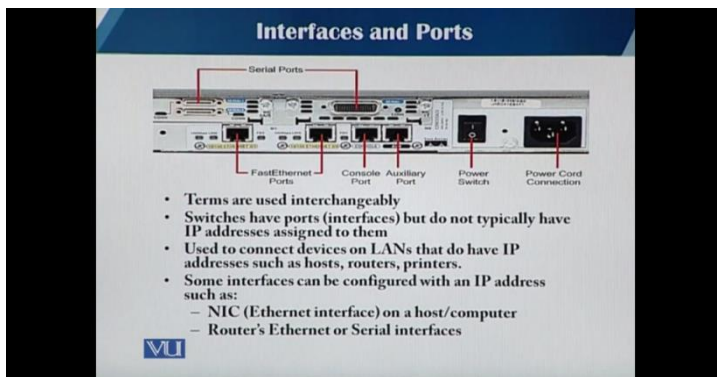
IP addressing of devices:



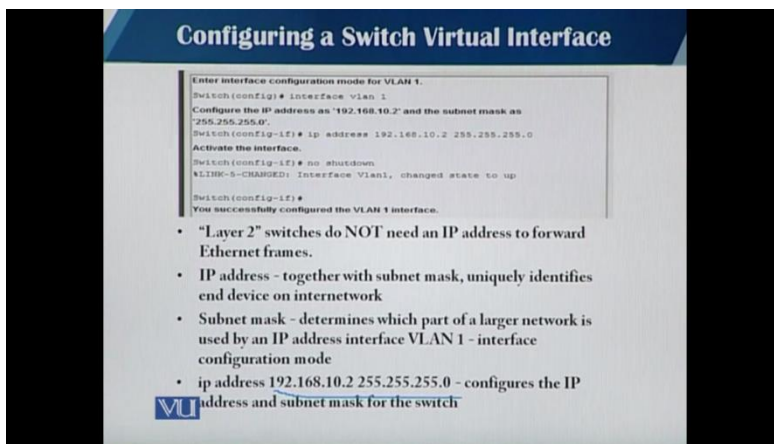
- Each device on a network must be known or configured via an IP address.
- Structure of IPv4 is also called dotted decimal.
- 192.168.10.1 is an example of IP address.
- Subnet Mask like 255.255.255.1 is also necessary with IP address.
- IP address can be assigned to both physical port and virtual port.

Interfaces & Ports:

Ip addresses are not assigned to the switches PORTS normally. Some of the characteristics of the ports are mentioned in the image below.

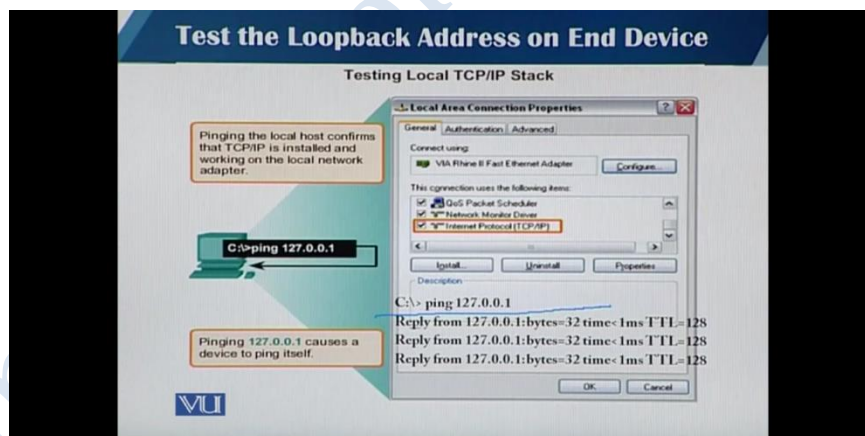


Configuring a Virtual Interface Switch



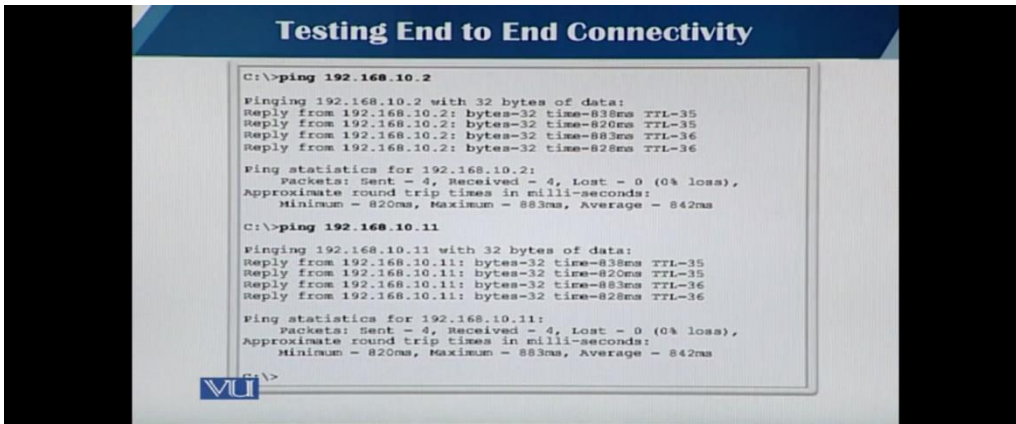
15. TOPIC

Loopback Address



- Pinging the local host confirms that TCP/IP is installed properly and is working fine on the local network.
- Ping command sends sequence of packets from one end to another to verify the sequence

Test end to end connectivity of a machine



16. Topic

Connecting to the Network

Whether connecting to a local printer in the home or to a web site in another country, before any network communications can occur, a physical connection to a local network must be established first. A physical connection can be a wired connection using a cable or a wireless connection using radio waves.

The type of physical connection used is totally dependent upon the setup of the network. For example, in many corporate offices, employees have desktop or laptop computers that are physically connected, via cable, to a shared switch. In addition to wired connections, some businesses may also offer wireless connections for laptops, tablets, and smartphones. With wireless devices, data is transmitted using radio waves. The use of wireless connectivity is becoming more common as individuals, and businesses alike, discover the advantages of offering wireless services. In order to offer wireless capability, a network must incorporate a wireless access point (WAP) for devices to connect to.

Switch devices and wireless access points are often two separate dedicated devices within a network implementation. However, there are also devices that offer both wired and wireless connectivity. In many homes, for example, individuals are implementing home integrated service routers (ISRs).

Network Interface Cards

Network Interface Cards (NICs) connect a device to the network. Ethernet NICs are used for a wired connection whereas WLAN (Wireless Local Area Network) NICs are used for wireless. An end-user device may include one or both types of NICs. A network printer, for example, may only have an Ethernet NIC, and therefore, must connect to the network using an Ethernet cable. Other devices, such as tablets and smartphones, might only contain a WLAN NIC and must use a wireless connection.

Physical Layer

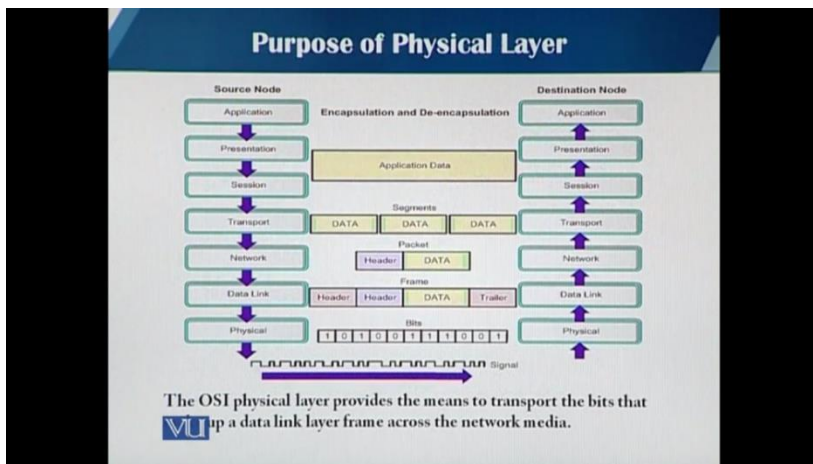
CS407

[Routing and switching]

[Reading material]

Made by Mah jabeen

[mahjabeen97869@gmail.com]



The OSI physical layer provides the means to transport the bits that make up a data link layer frame across the network media. This layer accepts a complete frame from the data link layer and encodes it as a series of signals that are transmitted onto the local media. The encoded bits that comprise a frame are received by either an end device or an intermediate device.

Forms of Network Media

There are three basic forms of network media.

- **Copper cable:** The signals are patterns of electrical pulses.
- **Fiber-optic cable:** The signals are patterns of light.
- **Wireless:** The signals are patterns of microwave transmissions.

Physical Layer Standards

The protocols and operations of the upper OSI layers are performed in software designed by software engineers and computer scientists. For example, the services and protocols in the TCP/IP suite are defined by the Internet Engineering Task Force (IETF) in RFCs

There are many different international and national organizations, regulatory government organizations, and private companies involved in establishing and maintaining physical layer standards. For instance, the physical layer hardware, media, encoding, and signaling standards are defined and governed by the:

- International Organization for Standardization (ISO)
- Telecommunications Industry Association/Electronic Industries Association (TIA/EIA)
- International Telecommunication Union (ITU)
- American National Standards Institute (ANSI)
- Institute of Electrical and Electronics Engineers (IEEE)
- National telecommunications regulatory authorities including the Federal Communication Commission (FCC) in the USA and the European Telecommunications Standards Institute (ETSI)

Physical Layer Bandwidth & Throughput

Bandwidth is the capacity of a medium to carry data. Digital bandwidth measures the amount of data that can flow from one place to another in a given amount of time. Bandwidth is typically measured in kilobits per second (kb/s) or megabits per second (Mb/s).

Throughput is the measure of the transfer of bits across the media over a given period of time.

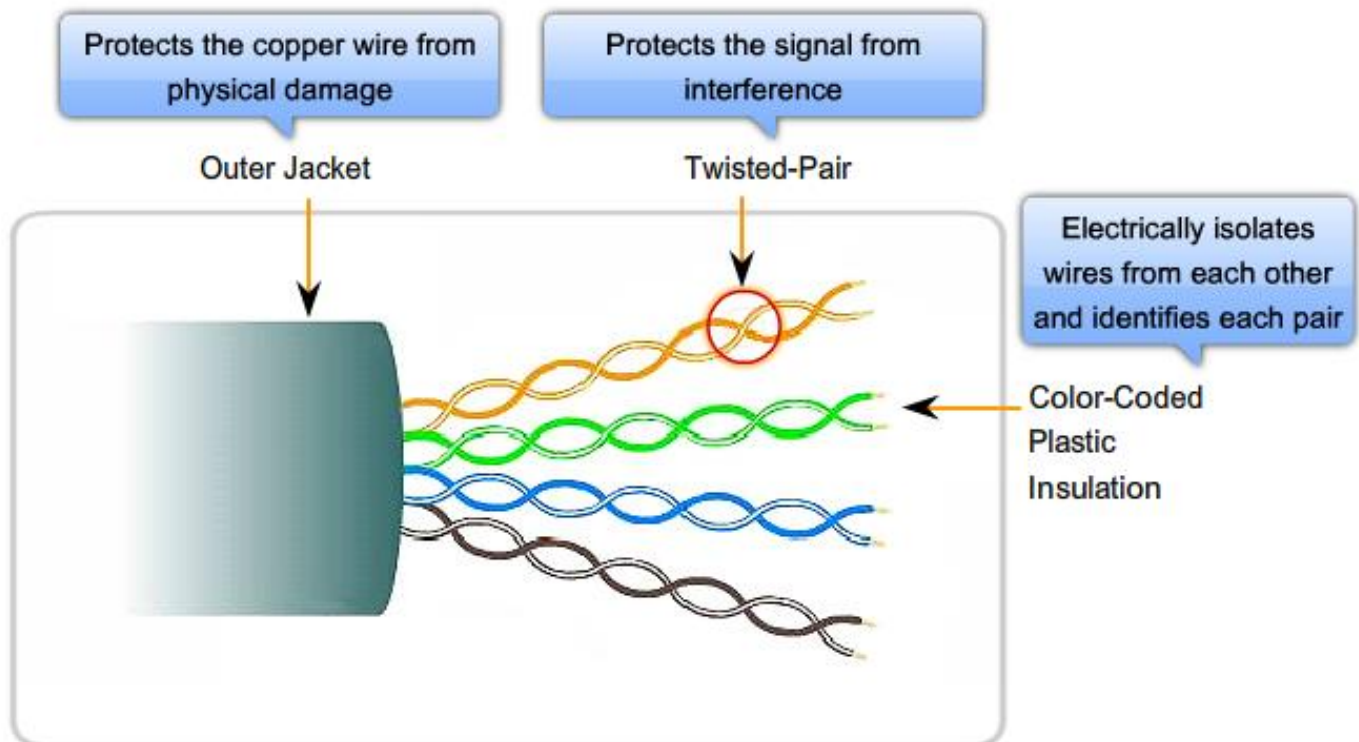
17. Topic

Network Media

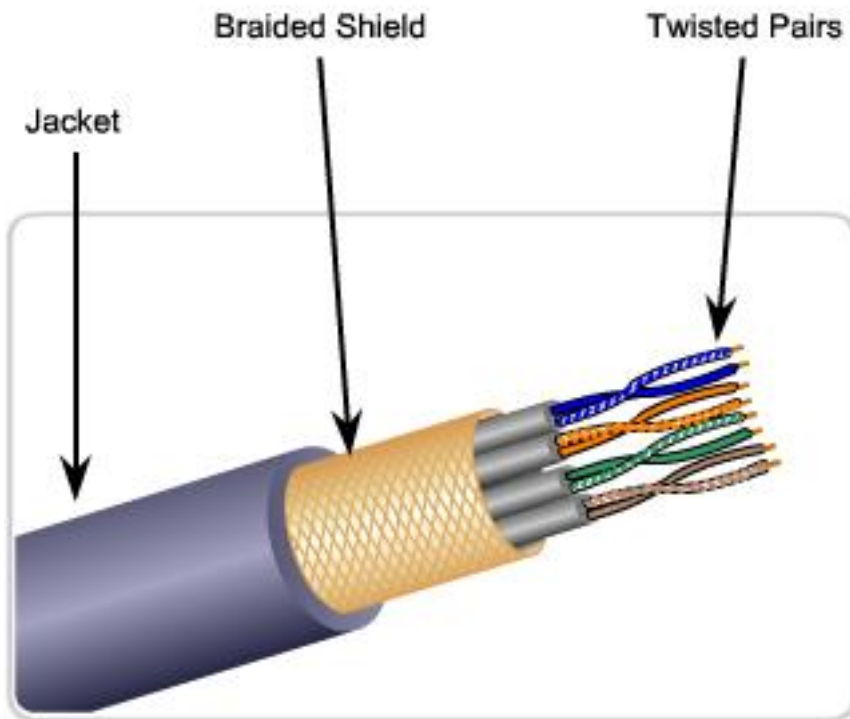
Unshielded twisted-pair (UTP)

UTP cabling is the most common networking media. UTP cabling, terminated with RJ-45 connectors, is used for interconnecting network hosts with intermediate networking devices, such as switches and routers.

In LANs, UTP cable consists of four pairs of color-coded wires that have been twisted together and then encased in a flexible plastic sheath which protects from minor physical damage. The twisting of wires helps protect against signal interference from other wires.



Shielded twisted pair (STP)



Shielded twisted-pair (STP) provides better noise protection than UTP cabling. However, compared to UTP cable, STP cable is significantly more expensive and difficult to install. Like UTP cable, STP uses an RJ-45 connector.

STP cable combines the techniques of shielding to counter EMI and RFI and wire twisting to counter crosstalk. To gain the full benefit of the shielding, STP cables are terminated with special shielded STP data connectors. If the cable is improperly grounded, the shield may act like an antenna and pick up unwanted signals.

Coaxial Cable

It consist of:

- A copper conductor used to transmit the electronic signals.
- The copper conductor is surrounded by a layer of flexible plastic insulation.
- The insulating material is surrounded in a woven copper braid, or metallic foil, that acts as the second wire in the circuit and as a shield for the inner conductor. This second layer, or shield, also reduces the amount of outside electromagnetic interference.
- The entire cable is covered with a cable jacket to protect it from minor physical damage.

Fiber Optic Cable

Optical fiber cable has become very popular for interconnecting infrastructure network devices. It permits the transmission of data over longer distances and at higher bandwidths (data rates) than any other networking media because data travels in the form of light.

Optical fiber is a flexible but extremely thin transparent strand of very pure glass (silica) not much bigger than a human hair. Bits are encoded on the fiber as light impulses.

Standard	Maximum Speed	Frequency	Compatible
802.11a	54 Mbps	5 GHz	No

802.11b	11 Mbps	2.4 GHz	No
802.11g	54 Mbps	2.4 GHz	802.11b
802.11n	600 Mbps	2.4 GHz or 5 GHz	802.11b/g
802.11ac	1.3 Gbps (1300 Mbps)	2.4 GHz and 5.5 GHz	802.11b/g/n

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Data Link Layer Protocols

The data link layer is responsible for the exchange of frames between nodes over a physical network media. It allows the upper layers to access the media and controls how data is placed and received on the media.

Specifically the data link layer performs these two basic services:

- It accepts Layer 3 packets and packages them into data units called frames.
- It controls media access control and performs error detection.

Data link Sublayers

The data link layer is actually divided into two sub layers:

- **Logical Link Control (LLC):**
This upper sub layer defines the software processes that provide services to the network layer protocols. It places information in the frame that identifies which network layer protocol is being used for the frame. This information allows multiple Layer 3 protocols, such as IPv4 and IPv6, to utilize the same network interface and media.
- **Media Access Control (MAC):**
This lower sub layer defines the media access processes performed by the hardware. It provides data link layer addressing and delimiting of data according to the physical signaling requirements of the medium and the type of data link layer protocol in use.

Data link Frame Header & Trailer

The frame header contains the control information specified by the data link layer protocol for the specific logical topology and media used.

Data link layer protocols add a trailer to the end of each frame. The trailer is used to determine if the frame arrived without error. This process is called error detection and is accomplished by placing a logical or mathematical summary of the bits that comprise the frame in the trailer.

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Data Link Layer Frame Fields

Header Fields:

- **Start Frame field:** Indicates the beginning of the frame.

- **Source and Destination Address fields:** Indicates the source and destination nodes on the media.
- **Type field:** Indicates the upper layer service contained in the frame.

Different data link layer protocols may use different fields from those mentioned. For example other Layer 2 protocol header frame fields could include:

- **Priority/Quality of Service field:** Indicates a particular type of communication service for processing.
- **Logical connection control field:** Used to establish a logical connection between nodes.
- **Physical link control field:** Used to establish the media link.
- **Flow control field:** Used to start and stop traffic over the media.
- **Congestion control field:** Indicates congestion in the media.

A transmitting node creates a logical summary of the contents of the frame. This is known as the cyclic redundancy check (CRC) value. This value is placed in the Frame Check Sequence (FCS) field of the frame to represent the contents of the frame.

xx. Click the FCS and Stop Frame fields in the figure for more information.

xx. When the frame arrives at the destination node, the receiving node calculates its own logical summary, or CRC, of the frame. The receiving node compares the two CRC values. If the two values are the same, the frame is considered to have arrived as transmitted. If the CRC value in the FCS differs from the CRC calculated at the receiving node, the frame is discarded.

xx. Therefore, the FCS field is used to determine if errors occurred in the transmission and reception of the frame. The error detection mechanism provided by the use of the FCS field discovers most errors caused on the media.

Ethernet Protocols for LAN'S

Preamble - Used for synchronization; also contains a delimiter to mark the end of the timing information

Destination Address - 48-bit MAC address for the destination node

Source Address - 48-bit MAC address for the source node

Type - Value to indicate which upper layer protocol will receive the data after the Ethernet process is complete

Data or payload - This is the PDU, typically an IPv4 packet, that is to be transported over the media.

Frame Check Sequence (FCS) - A value used to check for damaged frames

P2P Protocol for WAN'S

Flag - A single byte that indicates the beginning or end of a frame. The flag field consists of the binary sequence 01111110.

Address - A single byte that contains the standard PPP broadcast address. PPP does not assign individual station addresses.

Control - A single byte that contains the binary sequence 00000011, which calls for transmission of user data in an unsequenced frame.

Protocol - Two bytes that identify the protocol encapsulated in the data field of the frame. The most up-to-date values of the protocol field are specified in the most recent Assigned Numbers Request For Comments (RFC).
Data - Zero or more bytes that contain the datagram for the protocol specified in the protocol field.

Frame Check Sequence (FCS) - Normally 16 bits (2 bytes). By prior agreement, consenting PPP implementations can use a 32-bit (4-byte) FCS for improved error detection.

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Network Layer

The network layer, or OSI Layer 3, provides services to allow end devices to exchange data across the network. To accomplish this end-to-end transport, the network layer uses four basic processes:

Addressing end devices:

In the same way that a phone has a unique telephone number, end devices must be configured with a unique IP address for identification on the network. An end device with a configured IP address is referred to as a host.

Encapsulation

The network layer receives a protocol data unit (PDU) from the transport layer. In a process called encapsulation, the network layer adds IP header information, such as the IP address of the source (sending) and destination (receiving) hosts. After header information is added to the PDU, the PDU is called a packet.

Routing

The network layer provides services to direct packets to a destination host on another network. To travel to other networks, the packet must be processed by a router. The role of the router is to select paths for and direct packets toward the destination host in a process known as routing. A packet may cross many intermediary devices before reaching the destination host. Each route the packet takes to reach the destination host is called a hop.

De-encapsulation

When the packet arrives at the network layer of the destination host, the host checks the IP header of the packet. If the destination IP address within the header matches its own IP address, the IP header is removed from the packet. This process of removing headers from lower layers is known as de-encapsulation. After the packet is de-encapsulated by the network layer, the resulting Layer 4 PDU is passed up to the appropriate service at the transport layer.

Common Protocols

There are several network layer protocols in existence; however, only the following two are commonly implemented:

- Internet Protocol version 4 (IPv4)
- Internet Protocol version 6 (IPv6)

Other legacy network layer protocols that are not widely used include:

- Novell Internetwork Packet Exchange (IPX)
- AppleTalk
- Connectionless Network Service

Characteristics

- **Connectionless**
No connection with the destination is established before sending data packets.
- **Best Effort (unreliable)**

Packet delivery is not guaranteed.

- **Media Independent**

Operation is independent of the medium carrying the data.

Media Independent

The network layer is also not burdened with the characteristics of the media on which packets are transported. IP operates independently of the media that carry the data at lower layers of the protocol stack.

21. TOPIC

IPv4 Packet Structure

An IPv4 packet has two parts:

- **IP Header** - Identifies the packet characteristics.
- **Payload** - Contains the Layer 4 segment information and the actual data.

An IPv4 packet header consists of fields containing important information about the packet. These fields contain binary numbers which are examined by the Layer 3 process. The binary values of each field identify various settings of the IP packet.

Significant fields in the IPv4 header include:

- **Version**
Contains a 4-bit binary value identifying the IP packet version. For IPv4 packets, this field is always set to 0100.

- **Differentiated Services (DS)**

Formerly called the Type of Service (ToS) field, the DS field is an 8-bit field used to determine the priority of each packet. The first 6 bits identify the Differentiated Services Code Point (DSCP) value that is used by a quality of service (QoS) mechanism. The last 2 bits identify the explicit congestion notification (ECN) value that can be used to prevent dropped packets during times of network congestion.

- **Time-to-Live (TTL)**

Contains an 8-bit binary value that is used to limit the lifetime of a packet. It is specified in seconds but is commonly referred to as hop count. The packet sender sets the initial time-to-live (TTL) value and is decreased by one each time the packet is processed by a router, or hop. If the TTL field decrements to zero, the router discards the packet and sends an Internet Control Message Protocol (ICMP) Time Exceeded message to the source IP address. The traceroute command uses this field to identify the routers used between the source and destination.

- **Protocol**

This 8-bit binary value indicates the data payload type that the packet is carrying, which enables the network layer to pass the data to the appropriate upper-layer protocol. Common values include ICMP (0x01), TCP (0x06), and UDP (0x11).

- **Source IP Address**

Contains a 32-bit binary value that represents the source IP address of the packet.

- **Destination IP Address**

Contains a 32-bit binary value that represents the destination IP address of the packet.

- **Header CheckSum**

It is of 16 bits Field is used for error checking of the IP header. The checksum of the header is recalculated and compared to the value in the checksum field. If the values do not match, the packet is discarded.

- **Fragment Offset (13 bits)**

Field identifies the order in which to place the packet fragment in the reconstruction of the original unfragmented packet.

- **Flag (3 bits)**

This 3-bit field identifies how the packet is fragmented. It is used with the Fragment Offset and Identification fields to help reconstruct the fragment into the original packet.

- **Total Length (16 bits)**

Sometimes referred to as the Packet Length. Defines the entire packet (fragment) size, including header and data, in bytes. The minimum length packet is 20 bytes (20-byte header + 0 bytes data) and the maximum is 65,535 bytes.

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IPv4 Address Structure

IP address is a hierarchical address that is made up of two parts:

- Network portion
- Host portion

The bits within the network portion of the address must be identical for all devices that reside in the same network. The bits within the host portion of the address must be unique to identify a specific host within a network. Regardless of whether the decimal numbers between two IPv4 addresses match up, if two hosts have the same bit-pattern in the specified network portion of the 32-bit stream, those two hosts will reside in the same network.

IPv4 Subnet Mask

When an IPv4 address is assigned to a device, that device uses the subnet mask to determine what network address the device belongs to. The network address is the address that represents all the devices on the same network.

When sending network data, the device uses this information to determine whether it can send packets locally, or if it must send the packets to a default gateway for remote delivery. When a host sends a packet, it compares the network portion of its own IP address to the network portion of the destination IP address, based on subnet masks. If the network bits match, both the source and destination host are on the same network and the packet can be delivered locally. If they do not match, the sending host forwards the packet to the default gateway to be sent on to the other network.

Types of Addresses

There are three types of addresses within the address range of each IPv4 network:

- Network address
- Host addresses
- Broadcast address

Network Address

The network address is a standard way to refer to a network. The subnet mask or the prefix length might also be used when referring to network address.

Host Address

Every end device requires a unique address to communicate on the network. In IPv4 addresses, the values between the network address and the broadcast address can be assigned to end devices in a network

Broadcast Address

The IPv4 broadcast address is a special address for each network that allows communication to all the hosts in that network. To send data to all hosts in a network at once, a host can send a single packet that is addressed to the broadcast address of the network, and each host in the network that receives this packet will process its contents.

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IPv4 Addresses

Hosts are assigned IP addresses from the range of available addresses in the network. These IP addresses can be assigned either:

- **Statically**
- **Dynamically**

Source IP Addresses are always unicast

- **Unicasts:**

Packet travels from one host to another specific host. Unicast communication is used for normal host-to-host communication in both a client/server and a peer-to-peer network. Unicast packets use the addresses of the destination device as the destination address and can be routed through an internetwork.

- **Multicasts:**

Packet travels from one host to a select number of other hosts. Supports voice and audio broadcasts, news feeds.

Multicast transmission is designed to conserve the bandwidth of an IPv4 network. It reduces traffic by allowing a host to send a single packet to a selected set of hosts that are part of a subscribing multicast group. To reach multiple destination hosts using unicast communication, a source host would need to send an individual packet addressed to each host. With multicast, the source host can send a single packet that can reach thousands of destination hosts. The internetwork responsibility is to replicate the multicast flows in an efficient manner so that they reach only their intended recipients.

Examples of multicast transmission are: Video and audio broadcasts, Routing information exchange by routing protocols, Distribution of software

- **Broadcasts:**

Packet travels from one host to all hosts on the local network.

Broadcast traffic is used to send packets to all hosts in the network using the broadcast address for the network. With a broadcast, the packet contains a destination IP address with all ones (1s) in the host portion. This means that all hosts on that local network (broadcast domain) will receive and look at the packet. Many network protocols, such as DHCP, use broadcasts. When a host receives a packet sent to the network broadcast address, the host processes the packet as it would a packet addressed to its unicast address.

Some examples for using broadcast transmission are: Mapping upper layer addresses to lower layer addresses, requesting an address.

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IPv4 Address Types

IPv4 addresses are of two types:

- **Private Address**
- **Public Address**

Most IPv4 addresses are public IP addresses. These are reachable IPv4 addresses on the Internet.

However, there are blocks of addresses that are private addresses and are never propagated on the Internet. Packets with a **source or destination private IP address are not propagated by Internet routers**. Internet routers / firewalls block or translate these addresses.

Private addresses are usually converted to public IP addresses using **NAT (Network Address Translation)**. Private addresses are defined in RFC 1918.

The private address blocks are:

- - **10.0.0.0 /8:** 10.0.0.0 to 10.255.255.255
 - **172.16.0.0 /12:** 172.16.0.0 to 172.31.255.255
 - **192.168.0.0 /16:** 192.168.0.0 to 192.168.255.255

Special Use IPv4 Address

There are certain addresses that cannot be assigned to hosts. There are also special addresses that can be assigned to hosts, but with restrictions on how those hosts can interact within the network.

Network and Broadcast Addresses

As explained earlier, within each network the first and last addresses cannot be assigned to hosts. These are the network address and the broadcast address, respectively.

Loopback

One such reserved address is the IPv4 loopback address 127.0.0.1. The loopback is a special address that hosts use to direct traffic to themselves. The loopback address creates a shortcut method for TCP/IP applications and services that run on the same device to communicate with one another. By using the loopback address instead of the assigned IPv4 host address, two services on the same host can bypass the lower layers of the TCP/IP stack. You can also ping the loopback address to test the configuration of TCP/IP on the local host.

Although only the single 127.0.0.1 address is used, addresses 127.0.0.0 to 127.255.255.255 are reserved. Any address within this block will loop back to the local host. No address within this block should ever appear on any network.

Link-Local Addresses

IPv4 addresses in the address block 169.254.0.0 to 169.254.255.255 (169.254.0.0/16) are designated as link-local addresses. These addresses can be automatically assigned to the local host by the operating system in environments where no IP configuration is available. These might be used in a small peer-to-peer network or for a host that could not automatically obtain an address from a DHCP server.

Communication using IPv4 link-local addresses is only suitable for communication with other devices connected to the same network, as shown in the figure. A host must not send a packet with an IPv4 link-local destination address to any router for forwarding and should set the IPv4 time to live (TTL) for these packets to 1.

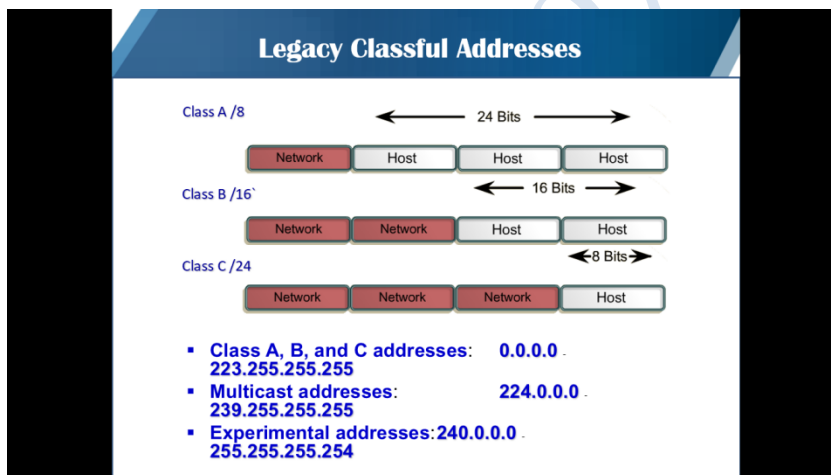
Link-local addresses do not provide services outside of the local network. However, many client/server and peer-to-peer applications will work properly with IPv4 link-local addresses.

TEST-NET Addresses

The address block 192.0.2.0 to 192.0.2.255 (192.0.2.0/24) is set aside for teaching and learning purposes. These addresses can be used in documentation and network examples. Unlike the experimental addresses, network devices will accept these addresses in their configurations. You may often find these addresses used with the domain names example.com or example.net in RFCs, vendor, and protocol documentation. Addresses within this block should not appear on the Internet.

Legacy Classful Address

RFC1700, Assigned Numbers, grouped the unicast ranges into specific sizes called class A, class B, and class C addresses. It also defined class D (multicast) and class E (experimental) addresses, as previously presented. The unicast address classes A, B, and C defined specifically-sized networks and specific address blocks for these networks. A company or organization was assigned an entire network from class A, class B, or class C address block. This use of address space is referred to as classful addressing.



Assignment of IP Address

Internet Assigned Numbers Authority (IANA) manages the allocation of IPv4 and IPv6 addresses. IPv4 address space are allocated to various other registries to manage for particular purposes or for regional areas. These registration companies are called Regional Internet Registries (RIRs).

25. TOPIC

Network Segmentation

Flat network

In early network implementations, it was common for organizations to have all computers and other networked devices connected to a single IP network. All devices in the organization were assigned an IP address with a matching network ID. This type of configuration is known as a flat network design.

In large networks, a flat network configuration creates major issues.

- Excessive broadcast traffic (e.g., DHCP, ARP) in one domain.
- Manageability and security

Subnetting

The process of segmenting a network, by dividing it into multiple smaller network spaces, is called subnetting. These sub-networks are called subnets. Network administrators can group devices and services into subnets that are determined by geographic location (perhaps the 3rd floor of a building), by organizational unit (perhaps the sales department), by device type (printers, servers, WAN), or any other division that makes sense for the network. Subnetting can reduce overall network traffic and improve network performance.

Reasons for Subnetting

- Create smaller broadcast domains.
- Limit the amount of traffic on the other network segments.
- Provide low-level security.

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Basic Subnetting

- Every network address has a valid range of host addresses. All devices attached to the same network will have an IPv4 host address for that network and a common subnet mask or network prefix.
- The prefix and the subnet mask are different ways of representing the same thing - the network portion of an address.
- IPv4 subnets are created by using one or more of the host bits as network bits. This is done by extending the mask to borrow some of the bits from the host portion of the address to create additional network bits. The more host bits borrowed, the more subnets that can be defined. For each bit borrowed, the number of subnetworks available is doubled. For example, if 1 bit is borrowed, 2 subnets can be created. If 2 bits, 4 subnets are created, if 3 bits are borrowed, 8 subnets are created, and so on. However, with each bit borrowed, fewer host addresses are available per subnet.
- Bits can only be borrowed from the host portion of the address. The network portion of the address is allocated by the service provider and cannot be changed.

Note: In the examples in the figures, only the last octet is shown in binary because only bits from the host portion can be borrowed.

As shown in Figure below, the 192.168.1.0/24 network has 24 bits in the network portion and 8 bits in the host portion, which is indicated with the subnet mask 255.255.255.0 or /24 notation. With no subnetting, this network supports a single LAN interface. If an additional LAN is needed, the network would need to be subnetted.

192.168.1.0/24 Network

Address	192	168	1	0000	0000
Mask	255	255	255	0000	0000
	Network Portion			Host Portion	

With no host bits borrowed, the host portion of both the network address and mask are all 0 bits.

In the below figure, 1 bit is borrowed from the most significant bit (leftmost bit) in the host portion, thus extending the network portion to 25 bits. This creates 2 subnets identified by using a 0 in the borrowed bit for the first network and a 1 in the borrowed bit for the second network. The subnet mask for both networks uses a 1 in the borrowed bit position to indicate that this bit is now part of the network portion.

Borrow 1 bit from the host portion of the address.

Original	192.	168.	1.	0	000	0000	1 Network
Mask	255.	255.	255.	0	000	0000	

The borrowed bit value is **0** for the Net 0 address.

Net 0	192.	168.	1.	0	000	0000	2 Subnets
Net 1	192.	168.	1.	1	000	0000	

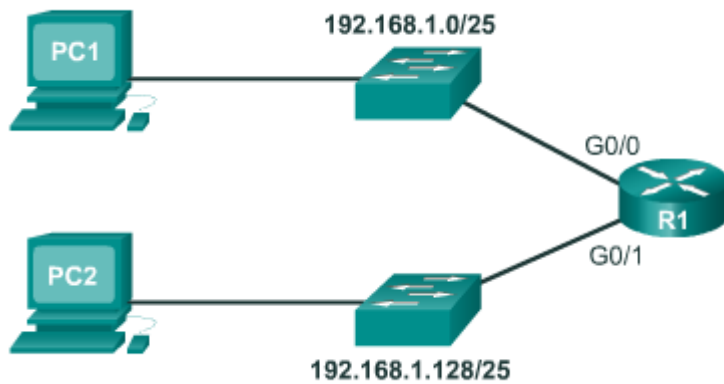
The new subnets have the **SAME** subnet mask.

Mask	255.	255.	255.	1	000	0000
------	------	------	------	---	-----	------

decimal.

At the end we convert the binary octet to

Subnets in use



In the previous example, the 192.168.1.0/24

network was subnetted to create two subnets:

- 192.168.1.0/25
- 192.168.1.128/25

In the above figure of subnetting, we notice that router R1 has two LAN segments attached to its GigabitEthernet interfaces. The subnets will be used for the segments attached to these interfaces. To serve as the gateway for devices on the LAN, each of the router interfaces must be assigned an IP address within the range of valid addresses for the assigned subnet. It is common practice to use the first or last available address in a network range for the router interface address.

The first subnet, 192.168.1.0/25, is used for the network attached to GigabitEthernet 0/0 and the second subnet, 192.168.1.128/25, is used for the network attached to GigabitEthernet 0/1. To assign an IP address for each of these interfaces, it is necessary to determine the range of valid IP addresses for each subnet.

The following are guidelines for each of the subnets:

- Network address - All 0 bits in the host portion of the address
- First host address - All 0 bits plus a right-most 1 bit in the host portion of the address.
- Last host address - All 1 bits plus a right-most 0 bit in the host portion of the address.
- Broadcast address - All 1 bits in the host portion of the address.

Subnetting Formulas

- **To calculate the number of subnets**

$$2^n \text{ (where } n = \text{the number of bits borrowed)}$$

- **To calculate the number of hosts**

$$2^n \text{ (where } n = \text{the number of bits remaining in the host field)}$$

27. TOPIC

Old/Traditional Subnetting

In traditional subnetting, each subnet is of fixed size and all accommodate the same no of hosts. If all the subnets possess the same requirement for the number of hosts then the traditional or fixed length subnetting is a better choice. Therefore in order to meet the variable number of hosts VLSM is considered to be the good choice.

VLSM Subnetting

- VLSM allows a network space to be divided in unequal parts.
- With VLSM the subnet mask will vary depending on how many bits have been borrowed for a particular subnet, thus the “variable” part of the VLSM.
- VLSM enables a network number to be configured with different subnet masks on different interfaces.
- Allows for more hierarchical levels within an addressing plan.
- Allows for better route summarization.

Cybersecurity in the context of Routing and Switching involves protecting networks from unauthorized access, data breaches, and malicious attacks.

It ensures that routers and switches are configured securely, use strong authentication methods, and employ encryption to safeguard data transmission.

Cybercrime in this context refers to illegal activities targeting routers and switches, such as hacking into networks, launching denial-of-service (DoS) attacks, or intercepting sensitive information.

These actions can disrupt network operations, compromise data integrity, and compromise network security.

28. TOPIC

Functions & Components of Router

Common functions of router are:

- Routers interconnects networks
- Routers choose best paths

Components of Routers includes:

- Operating systems (OS)
- Central processing units (CPU)
- Random-access memory (RAM)
- Read-only memory (ROM)
- Routers also have special memory that includes
- Flash
- Nonvolatile random-access memory (NVRAM).

Memory of Router

A router has access to four types of memory: RAM, ROM, NVRAM, and Flash.

RAM

RAM is used to store various applications and processes including:

- Cisco IOS - The IOS is copied into RAM during bootup. Running configuration file - This is the configuration file that stores the configuration commands that the router IOS is currently using. It is also known as the running-config.
- IP routing table - This file stores information about directly-connected and remote networks. It is used to determine the best path to use to forward packets.
- ARP cache - This cache contains the IPv4 address to MAC address mappings, similar to the Address Resolution Protocol (ARP) cache on a PC. The ARP cache is used on routers that have LAN interfaces, such as Ethernet interfaces.
- Packet buffer - Packets are temporarily stored in a buffer when received on an interface or before they exit an interface.

ROM

Cisco routers use ROM to store:

- Bootup instructions - Provides the startup instructions.
- Basic diagnostic software - Performs the power-on self-test (POST) of all components.
- Limited IOS - Provides a limited backup version of the OS, in case the router cannot load the full featured IOS.
- ROM is firmware embedded on an integrated circuit inside the router and does not lose its contents when the router loses power or is restarted.

NVRAM

NVRAM is used by the Cisco IOS as permanent storage for the startup configuration file (startup-config). Like ROM, NVRAM does not lose its contents when power is turned off.

Flash Memory

Flash memory is non-volatile computer memory used as permanent storage for the IOS and other system related files. The IOS is copied from flash into RAM during the bootup process.

Connections/Ports of Router:

A Cisco 1941 router includes the following connections:

- **Console ports** - Two console ports for the initial configuration and command-line interface (CLI) management access using a regular RJ-45 port and a new USB Type-B (mini-B USB) connector.
- **AUX port** - An RJ-45 port for remote management access; this is similar to the Console port.
- **Two LAN interfaces** - Two Gigabit Ethernet interfaces for LAN access.
- Enhanced high-speed WAN interface card (**EHWIC**) slots - Two slots that provide modularity and flexibility by enabling the router to support different types of interface modules, including Serial, digital subscriber line (DSL), switch port, and wireless.

Router Interfaces:

A router interface is a physical connector that enables a router to send or receive packets.

Types of router interfaces:

- Ethernet
- FastEthernet
- Gigabit Ethernet
- Serial

- DSL
- Cable
- ISDN

Router interfaces can be grouped into two categories:

- **Ethernet LAN interfaces:** Requires an IP address and enabled. Used for connecting cables that terminate with LAN devices, such as computers and switches. This interface can also be used to connect routers to each other. Several conventions for naming Ethernet interfaces are popular: the older Ethernet, FastEthernet, and Gigabit Ethernet.
- **Serial WAN interfaces** – Requires an IP address and enabled. Used for connecting routers to external networks, usually over a larger geographical distance. Similar to LAN interfaces, each serial WAN interface has its own IP address and subnet mask, which identifies it as a member of a specific network.

29. TOPIC

When router is booted it loads following two files in the RAM:

- Cisco IOS:

It resides in flash memory.

It facilitates the basic functions of device hardware components.

- Startup-config:

It resides in NVRAM.

It contains the commands to initially configure the router.

Configuration changes are stored in it and immediately implemented by IOS.

There three major phases of Router Bootup Process:

1. POST (Power on Self-Test) and loads bootstrap program both files are residing in ROM
2. Locating and loading of Cisco IOS operating system.
3. Locating and loading of start config file and enter in setup mode

30. TOPIC

Configuring the router and switch is almost similar, for configuring a cisco router you have to perform the following with the help of commands:

1. Name the device
2. Secure Management Access

3. Configure a Banner

31. TOPIC

1. To verify the connectivity of router we perform following tasks by means of giving commands on the console:
2. Verify interface settings by show IP command
3. We can also filter the show command as per our need (i.e. to see specific results)
4. Commands that generate multiple screens of output are, by default, paused after 24 lines.
5. Terminal maintains the history of 200 commands.

32. TOPIC

Every router has two tables:

1. Routing table:
Which contains the IP addresses of connected devices.
2. ARP table/cache:
Which contains the MAC addresses of connected devices.

Whereas sending and receiving devices have only ARP cache/table.

If there are some of the routers in between two connected PCs they

Rebuild the frame when one router receives and forwards the packet.

33. TOPIC

Router determine the path of a packet on the basis of one of the following metrics:

- Hop count
- Bandwidth
- Delay
- Load
- Reliability

There are some protocols which use one or the combination of these metrics to determine the path. Some are as under:

1. Routing Information Protocol (RIP) - Hop count

2. Open Shortest Path First (OSPF) - Cisco's cost based on cumulative bandwidth from source to destination
3. Enhanced Interior Gateway Routing Protocol (EIGRP) - Bandwidth, delay, load, reliability

When a router has two or more paths to a destination with equal cost metrics, then the router forwards the packets using both paths equally. This is called **equal cost load balancing**.

Week no 4

34. TOPIC

Routing table is a file loaded in the RAM of router it have an entry for each device connected with a router it can be:

- Directly connected Router
- Remote Router
- Next hope association

Show IP route command shows all these connected devices to the user.

35. TOPIC

Using CISCO packet tracer software, we can configure and connect different devices with each other. We can expand the functionality of any device by adding components. If you want to add any component, then you have to switch off the device, otherwise it cannot be added. Do switch ON the device after adding any component.

On clicking any device, a window appears that have three different following tabs:

- Physical: It shows the physical view of the device.
- Config: It shows the configuration of the selected device.
- CLI: We can see the IP addresses by giving commands.

36. TOPIC

ICMP have is a network layer protocol it has number of uses such as:

- Informational messages (ping, trace-route)
 - Ping is used to verify the connectivity of any device in a network.
 - Trace-rout is used to find the path between sender and receiver.
- Error messages (network unreachable)

37. TOPIC

CISCO packet traces is a network simulator, that help students to understand the concepts of networking. On the top of the CISCO packet traces there is a toolbar. At the bottom left corner of the tool, there are a lot of networking devices (switch, hub, end devices and wires) are available. You can simply drag and drop any device for use. Configuration information is visible in a box when we hover our mouse over any active device. Whenever we connect two devices the link first turns red and after several seconds (25-30) when spanning tree protocol (STP) run successfully it is active and turns green.

38. TOPIC

Before using any device, we have to configure the device first, configuring means assigning the IP address to a device. For assigning an IP address we simply click on the device and go to the config tab. IP addresses are of two types:

- Static - remain same - in the video we configure static IP.
- Dynamic - assigned by DHCP when any device connects in a network it is changed every time.

39. TOPIC

In the same way we have configured other devices, we can configure the servers. Their server can run a bunch of services that we make active. Clients can access any service that is running on a client. The active layers of OSI models are also shown in the OSI model tab.

40. TOPIC

We can access the CLI of a device by entering the terminal of the device. There are different modes of a terminal:

- Console
- User
- Exec

All these modes have different privileges.

41. TOPIC

A switch can have following different ports:

- Fast ethernet ports --24
- Gigabit ethernet ports -- 2
- VLAN port -- 1
- Console ports --2

For configuration of any device we must be in global configuration mode (i.e. terminal)

42. TOPIC

You will first perform basic switch configurations. Then you will implement basic

connectivity by configuring IP addressing on switches and PCs by using **EXEC** mode using *enable* command. When the IP addressing configuration is complete, you will use various *show* commands to verify configurations and use the *ping* command to verify basic connectivity between devices.

43. TOPIC

It is the continuation of the previous topic, just the testing is done using the ping command.

44. TOPIC

Different types of transmission medium is required to connect different networking devices some detail is mentioned in the table below:

Devices to connect		Cable
Cloud	Router	Copper Straight-Through/Ethernet
Cloud	Cable Modem	Coaxial Cable
Router	Router	Serial DCE
Router	Server	Copper Cross-over
Router	Configuration Terminal	Console
Router	Switch	Fiber
Wireless Router	Cable Modem	Copper Straight-Through/Ethernet
Wireless Router	PC	Copper Straight-Through/Ethernet

Week no 5

45. TOPIC

By using different types of protocols in CISCO simulator, we can generate the following types of traffics:

- Unicast - Only one is receiver → It uses ICMP as a protocol
- Multicast - More than one receiver but not all → It uses EIGRP as a protocol
- Broadcast - All are receiver → It uses EIGRP as a protocol

Routers do not take part in broadcasting but switches do.

46. TOPIC

We can design an IP Addressing Scheme(subnetting scheme) based on the number of connected devices in a network and subnetwork. We can find these devices by identifying the number of LANs and then counting the number of devices connected in any LAN.

47. TOPIC

It is a common practice that we assign the first address to the router while designing IP addresses in a LAN.

We can configure the devices with the help of commands and can check the connectivity with the ping command.

48. TOPIC

This topic including video title that is exactly similar to the topic no 35

49. TOPIC

One PC is combined with a router using console cable.

All the commands are exactly similar to the “configuring of switch in topic 41”

After the configuration is complete we must save the configuration in the startup file by using appropriate commands so that it will be saved in NV-RAM.

50. TOPIC

There are the following ways through which a router can learn about remote networks:

- Statically/Manually - Network administrator himself configure the router
- Dynamically - A routing protocol is used which configure the router

Some of the differences are shown in the table below:

CS407

[Routing and switching]

[Reading material]

Made by Mah jabeen

[mahjabeen97869@gmail.com]

	Dynamic Routing	Static Routing
Configuration Complexity	Generally independent of the network size	Increases with network size
Topology Changes	Automatically adapts to topology changes	Administrator intervention required
Scaling	Suitable for simple and complex topologies	Suitable for simple topologies
Security	Less secure	More secure
Resource Usage	Uses CPU, memory, link bandwidth	No extra resources needed
Predictability	Route depends on the current topology	Route to destination is always the same

We use static routes for the following reasons:

- In small networks that are not expected to grow significantly.
- To route traffic to and from stub networks.

A network which has only one route, from which it can be accessed is called stub network.

In stub networks we use static routes and that can be of the following types:

Standard static routes are useful when connecting to a specific remote network.

Default static routes are used when no other routes in the routing table match the packet destination IP address. In other words, when a more specific match does not exist. A common use is when connecting a company's edge router to the ISP network.

Summary static route

Floating static route

51. TOPIC

Summary static route reduce the number of routing table entries, multiple static routes can be summarized into a single static route if:

- The destination networks are contiguous and can be summarized into a single network address.
- The multiple static routes all use the same exit interface or next-hop IP address

Floating static route is also called a backup static route. It is used to provide a backup path to a primary static or dynamic route, in the event of a link failure. The floating static route is only used when the primary route is not available. If multiple paths to the destination exist, the router will choose the path with the lowest administrative distance.

52. TOPIC

ip route command is used to configure static and default routes some details are mentioned in following table:

ip route network-address subnet-mask {ip-address | interface-type interface-number [ip-address]} [distance] [name name] [permanent] [tag tag]

Parameter	Description
<i>network-add</i>	Destination network address of the remote network to be added to the routing table.
<i>subnet</i>	Subnet mask of the remote network to be added to the routing table. Note: The subnet mask can be modified to summarize a group of networks
<i>ip-address</i>	Commonly referred to as the next-hop router's IP address.
<i>exit-intf</i>	Use the outgoing interface to forward packets to the destination network. Also referred to as a directly attached static route. Typically used when connecting in a point-to-point configuration.
<i>distance</i>	Used to create a floating static route by setting an administrative distance that is higher than a dynamically learned route.

There are three standards static routes:

- **Next-hop route** - Only the next-hop IP address is specified.(CEF-CISCO Express Forwarding) used in it
- **Directly connected static route** - Only the router exit interface is specified.
- **Fully specified static route** - The next-hop IP address and exit interface are specified.

53. TOPIC

- **Directly connected static route** - Only the router exit interface is specified.
- This is an alternative method for configuring static routes on a point-to-point network.

- Static routes with a next-hop address recommended when CEF is enabled.
- **Fully specified static route** - Both the next-hop IP address and exit interface are specified.
- **Default static route** -
- The basic command syntax of a default static route is:
- `ip route 0.0.0.0 0.0.0.0 {ip-address | exit-intf }`

Parameter	Description
0.0.0.0	Matches any network address.
0.0.0.0	Matches any subnet mask.
<i>ip-address</i>	Commonly referred to as the next-hop router's IP address. Typically used when connecting to a broadcast media (i.e.Ethernet). Commonly creates a recursive lookup.
<i>exit-intf</i>	Use the outgoing interface to forward packets to the destination network. Also referred to as a directly attached static route. Typically used when connecting in a point-to-point configuration.

54. TOPIC

After successful configuration of any router the red dot on the wire turns green.

55. TOPIC

Configure Devices,Verify Connectivity and Gather Information with Show Commands

In this activity, you will configure a router with basic settings including IP addressing. You will also configure a switch for remote management and configure the PCs.

Week no 6

56. TOPIC

After you have successfully verified connectivity, you will use show commands to gather information about the network.

57. TOPIC

A router can learn about remote networks in one of two ways:

Manually - Remote networks are manually entered into the route table using static routes.

Dynamically - Remote routes are automatically learned using a dynamic routing protocol.

A network administrator can manually configure a static route to reach a specific network. Unlike a dynamic routing protocol, static routes are not automatically updated and must be manually reconfigured any time the network topology changes. A static route does not change until the administrator manually reconfigures it.

Dynamic vs. Static Routing

	Dynamic Routing	Static Routing
Configuration Complexity	Generally independent of the network size	Increases with network size
Topology Changes	Automatically adapts to topology changes	Administrator intervention required
Scaling	Suitable for simple and complex topologies	Suitable for simple topologies
Security	Less secure	More secure
Resource Usage	Uses CPU, memory, link bandwidth	No extra resources needed
Predictability	Route depends on the current topology	Route to destination is always the same

Static routing has three primary uses:

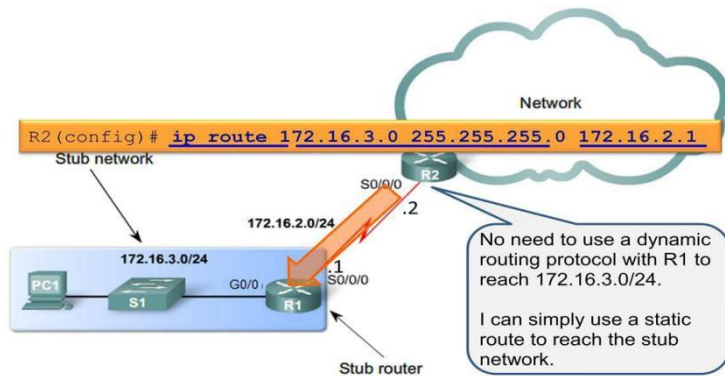
- Providing ease of routing table maintenance in smaller networks that are not expected to grow significantly.
- Routing to and from stub networks. A stub network is a network accessed by a single route, and the router has only one neighbor.
- Using a single default route to represent a path to any network that does not have a more specific match with another route in the routing table. Default routes are used to send traffic to any destination beyond the next upstream router.

Static routes can also be used to:

- Reduce the number of routes advertised by summarizing several contiguous networks as one static route
- Create a backup route in case a primary route link fails

The following types of IPv4 and IPv6 static routes will be discussed:

- Standard static route
- Default static route
- Summary static route
- Floating static route



A default static route is a route that matches all packets. A default route identifies the gateway IP address to which the router sends all IP packets that it does not have a learned or static route. A default static route is simply a static route with 0.0.0.0/0 as the destination IPv4 address. Configuring a default static route creates a Gateway of Last Resort.

Default static routes are used:

- When no other routes in the routing table match the packet destination IP address. In other words, when a more specific match does not exist. A common use is when connecting a company's edge router to the ISP network.
- When a router has only one other router to which it is connected. This condition is known as a stub router.

Summary Static Route

To reduce the number of routing table entries, multiple static routes can be summarized into a single static route if:

- The destination networks are contiguous and can be summarized into a single network address.
- The multiple static routes all use the same exit interface or next-hop IP address.

Floating Static Route

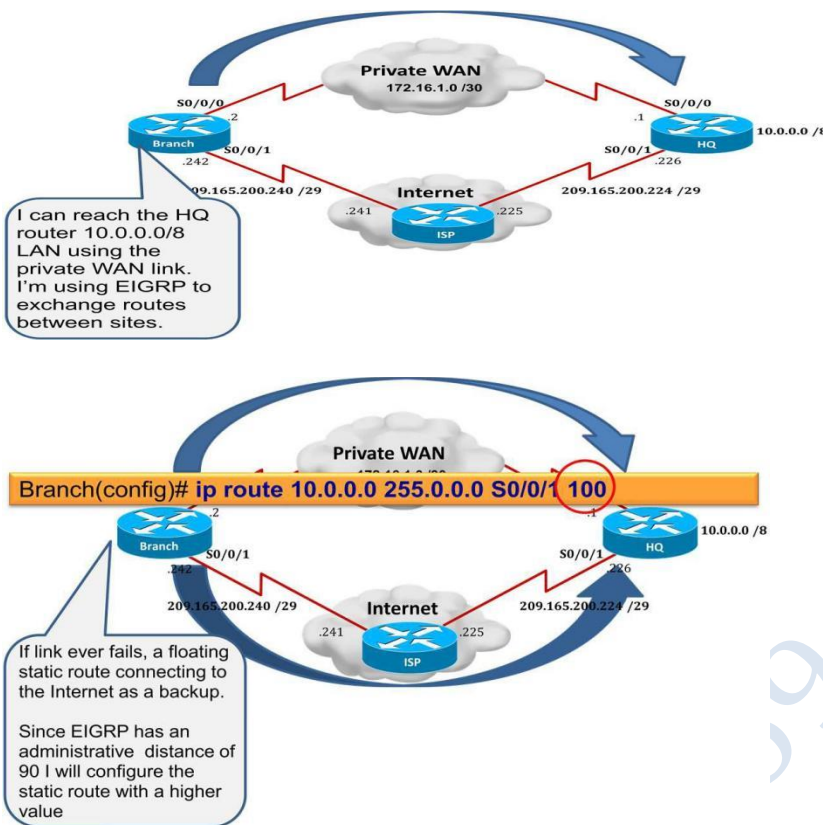
Another type of static route is a floating static route. Floating static routes are static routes that are used to provide a backup path to a primary static or dynamic route, in the event of a link failure. The floating static route is only used when the primary route is not available.

To accomplish this, the floating static route is configured with a higher administrative distance than the primary route. Recall that the administrative distance represents the trustworthiness of a route. If multiple paths to the destination exist, the router will choose the path with the lowest administrative distance.

For example, assume that an administrator wants to create a floating static route as a backup to an EIGRP-learned route. The floating static route must be configured with a higher administrative distance than EIGRP. EIGRP has an administrative distance of 90. If the floating static route is configured with an administrative distance of 95, the dynamic route learned through EIGRP is preferred to the floating static route. If the EIGRP-learned route is lost, the floating static route is used in its place.

In the figure below, the Branch router typically forwards all traffic to the HQ router over the private WAN link. In this example, the routers exchange route information using EIGRP. A floating static route, with an

administrative distance of 91 or higher, could be configured to serve as a backup route. If the private WAN link fails and the EIGRP route disappears from the routing table, the router selects the floating static route as the best path to reach the HQ LAN.



58. TOPIC

Configuring IPv4 Static & Default Routes

- Used to reduce the number of routing table entries.
- Multiple static routes can be summarized into a single static route if:
- The destination networks are contiguous
- The destination networks are all reachable using the same exit interface

To reduce the number of routing table entries, multiple static routes can be summarized into a single static route if:

- The destination networks are contiguous and can be summarized into a single network address.
- The multiple static routes all use the same exit interface or next-hop IP address.
- In the figure, R1 would require four separate static routes to reach the 172.20.0.0/16 to 172.23.0.0/16 networks. Instead, one summary static route can be configured and still provide connectivity to those networks.

Floating Static Route

Another type of static route is a floating static route. Floating static routes are static routes that are used to provide a backup path to a primary static or dynamic route, in the event of a link failure. The floating static route is only used when the primary route is not available.

To accomplish this, the floating static route is configured with a higher administrative distance than the primary route. Recall that the administrative distance represents the trustworthiness of a route. If multiple paths to the destination exist, the router will choose the path with the lowest administrative distance.

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In the figure, the Branch router typically forwards all traffic to the HQ router over the private WAN link. In this example, the routers exchange route information using EIGRP. A floating static route, with an administrative distance of 91 or higher, could be configured to serve as a backup route. If the private WAN link fails and the EIGRP route disappears from the routing table, the router selects the floating static route as the best path to reach the HQ LAN.

59. TOPIC

ip route *network-add subnet* {*ip-address* | *exit-intf* [*ip-address*]} [*distance*]

Parameter	Description
<i>network-add</i>	Destination network address of the remote network to be added to the routing table.
<i>subnet</i>	- Subnet mask of the remote network to be added to the routing table. - Note: The subnet mask can be modified to summarize a group of networks
<i>ip-address</i>	Commonly referred to as the next-hop router's IP address.
<i>exit-intf</i>	- Use the outgoing interface to forward packets to the destination network. - Also referred to as a directly attached static route. - Typically used when connecting in a point-to-point configuration.
<i>distance</i>	Used to create a floating static route by setting an administrative distance that is higher than a dynamically learned route.

Next Hop Static Route

In a next-hop static route, only the next-hop IP address is specified. The output interface is derived from the next hop.

Please see the slide for an example.

Directly Connected Static Route

When configuring a static route, another option is to use the exit interface to specify the next-hop address. In older IOS versions, prior to CEF, this method is used to avoid the recursive lookup problem.

Configuring a directly connected static route with an exit interface allows the routing table to resolve the exit interface in a single search, instead of two searches.

Note: For point-to-point interfaces, you can use static routes that point to the exit interface or to the next-hop address. For multipoint/broadcast interfaces, it is more suitable to use static routes that point to a next-hop address.

Please see the slide for an example.

Fully Specified Static Route

In a fully specified static route, both the output interface and the next-hop IP address are specified. This is another type of static route that is used in older IOSs, prior to CEF. This form of static route is used when the output interface is a multi-access interface and it is necessary to explicitly identify the next hop. The next hop must be directly connected to the specified exit interface.

Please see the slide for an example.

Verify a Static Route

```
R1# show ip route static | begin Gateway
```

```
Gateway of last resort is not set
```

```
172.16.0.0/16 is variably subnetted, 5 subnets, 2 masks
```

```
S    172.16.1.0/24 [1/0] via 172.16.2.2
```

```
S    192.168.1.0/24 [1/0] via 172.16.2.2
```

```
S    192.168.2.0/24 [1/0] via 172.16.2.2
```

```
R1# show ip route 192.168.2.1
```

```
Routing entry for 192.168.2.0/24
```

```
Known via "static", distance 1, metric 0
```

```
Routing Descriptor Blocks:
```

```
* 172.16.2.2
```

```
Route metric is 0, traffic share count is 1
```

```
R1# show running-config | section ip route
```

```
ip route 172.16.1.0 255.255.255.0 172.16.2.2
```

```
ip route 192.168.1.0 255.255.255.0 172.16.2.2
```

```
ip route 192.168.2.0 255.255.255.0 172.16.2.2
```

Default Static Route

```
ip route 0.0.0.0 0.0.0.0 {ip-address | exit-intf [ip-address]}
```

Parameter	Description
0.0.0.0	• Matches any network address.
0.0.0.0	• Matches any subnet mask.
<i>ip-address</i>	• Commonly referred to as the next-hop router's IP address. • Typically used when connecting to a broadcast media (i.e., Ethernet) . • Commonly creates a recursive lookup.
<i>exit-intf</i>	• Use the outgoing interface to forward packets to the destination network. • Also referred to as a directly attached static route. • Typically used when connecting in a point-to-point configuration.

Configure Default Static Route

A default route is a static route that matches all packets. Rather than storing all routes to all networks in the routing table, a router can store a single default route to represent any network that is not in the routing table.

Routers commonly use default routes that are either configured locally or learned from another router, using a dynamic routing protocol. A default route is used when no other routes in the routing table match the destination IP address of the packet. In other words, if a more specific match does not exist, then the default route is used as the Gateway of Last Resort.

Default static routes are commonly used when connecting:

- An edge router to a service provider network
- A stub router (a router with only one upstream neighbor router)

The command syntax for a default static route is similar to any other static route, except that the network address is **0.0.0.0** and the subnet mask is **0.0.0.0**. The basic command syntax of a default static route is:

```
ip route 0.0.0.0 0.0.0.0 {ip-address | exit-intf }
```

Verify Default Static Route

R1# show ip route static

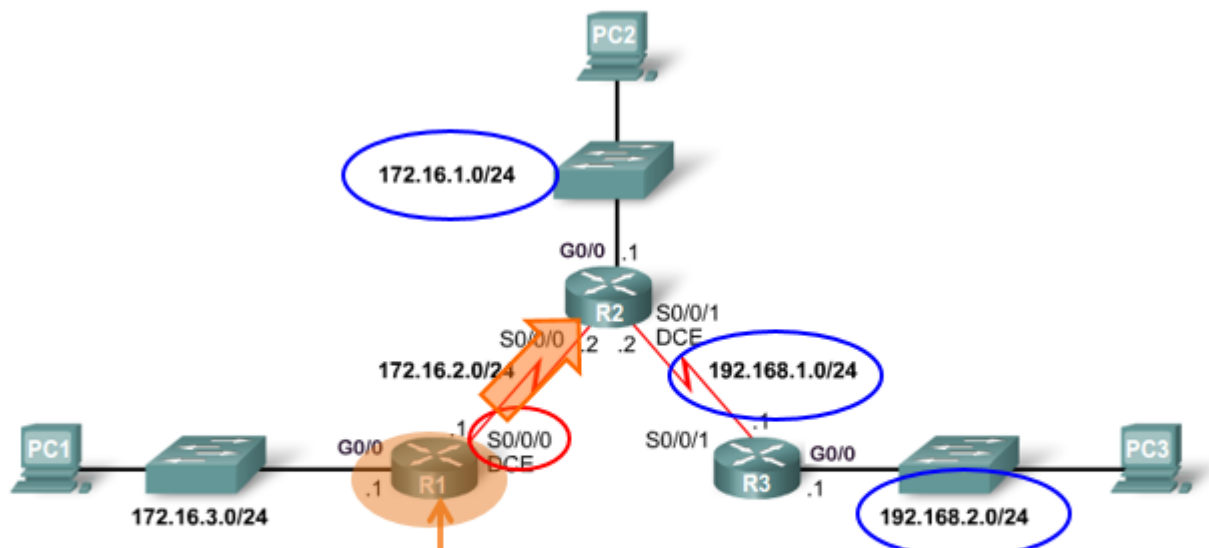
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, I - LISP
+ - replicated route, % - next hop override

Gateway of last resort is 0.0.0.0 to network 0.0.0.0

S* 0.0.0.0/0 is via 172.16.2.2

R1#

60. TOPIC



```

R1(config)# ip route 172.16.1.0 255.255.255.0 s0/0/0
R1(config)# ip route 192.168.1.0 255.255.255.0 s0/0/0
R1(config)# ip route 192.168.2.0 255.255.255.0 s0/0/0
R1(config)#

```

When configuring a static route, another option is to use the exit interface to specify the next-hop address. In older IOS versions, prior to CEF, this method is used to avoid the recursive lookup problem.

In Figure 1, three directly connected static routes are configured on R1 using the exit interface. The routing table for R1 in Figure 2 shows that when a packet is destined for the 192.168.2.0/24 network, R1 looks for a match in the routing table, and finds that it can forward the packet out of its Serial 0/0/0 interface. No other lookups are required.

Notice how the routing table looks different for the route configured with an exit interface than the route configured with a recursive entry.

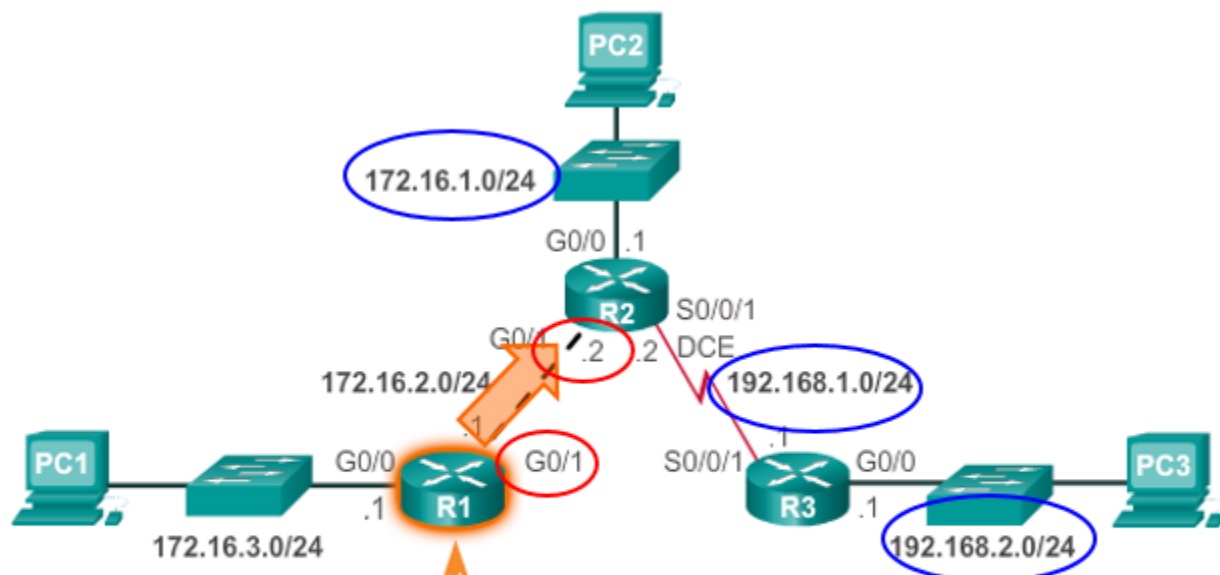
Configuring a directly connected static route with an exit interface allows the routing table to resolve the exit interface in a single search, instead of two searches. Although the routing table entry indicates directly connected, the administrative distance of the static route is still 1. Only a directly connected interface can have an administrative distance of 0.

For point-to-point interfaces, you can use static routes that point to the exit interface or to the next-hop address. For multipoint/broadcast interfaces, it is more suitable to use static routes that point to a next-hop address.

Use the Syntax Checker in Figures 3 and 4 to configure and verify directly connected static routes on R2 and R3.

Although static routes that use only an exit interface on point-to-point networks are common, the use of the default CEF forwarding mechanism makes this practice unnecessary.

Fully Specified Static Route



```
R1(config)# ip route 172.16.1.0 255.255.255.0 G0/1 172.16.2.2
R1(config)# ip route 192.168.1.0 255.255.255.0 G0/1 172.16.2.2
R1(config)# ip route 192.168.2.0 255.255.255.0 G0/1 172.16.2.2
R1(config)#
```

In a fully specified static route, both the output interface and the next-hop IP address are specified. This is another type of static route that is used in older IOSs, prior to CEF. This form of static route is used when the output interface is a multi-access interface and it is necessary to explicitly identify the next hop. The next hop must be directly connected to the specified exit interface.

Suppose that the network link between R1 and R2 is an Ethernet link and that the Gigabit Ethernet 0/1 interface of R1 is connected to that network, as shown in Figure 1. CEF is not enabled. To eliminate the recursive lookup, a directly connected static route can be implemented using the following command:

```
R1(config)# ip route 192.168.2.0 255.255.255.0 Gigabit Ethernet 0/1
```

However, this may cause unexpected or inconsistent results. The difference between an Ethernet multi-access network and a point-to-point serial network is that a point-to-point network has only one other device on that network, the router at the other end of the link. With Ethernet networks, there may be many different devices sharing the same multi-access network, including hosts and even multiple routers. By only designating the Ethernet exit interface in the static route, the router will not have sufficient information to determine which device is the next-hop device.

R1 knows that the packet needs to be encapsulated in an Ethernet frame and sent out the Gigabit Ethernet 0/1 interface. However, R1 does not know the next-hop IPv4 address and therefore it cannot determine the destination MAC address for the Ethernet frame.

Depending upon the topology and the configurations on other routers, this static route may or may not work. It is recommended that when the exit interface is an Ethernet network, that a fully specified static route is used including both the exit interface and the next-hop address.

As shown in Figure 2, when forwarding packets to R2, the exit interface is Gigabit Ethernet 0/1 and the next-hop IPv4 address is 172.16.2.2.

With the use of CEF, a fully specified static route is no longer necessary. A static route using a next-hop address should be used.

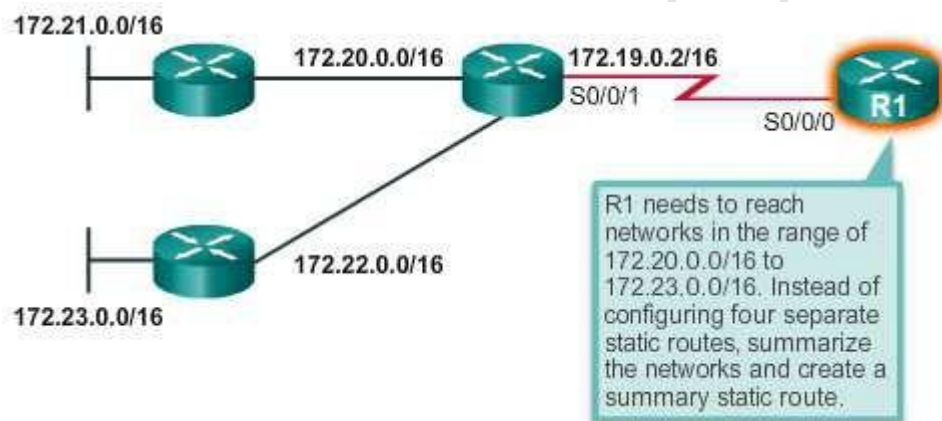
Use the Syntax Checker in Figure 3 and 4 to configure and verify fully specified static routes on R2 and R3.

61. TOPIC

Route summarization, also known as route aggregation, is the process of advertising a contiguous set of addresses as a single address with a less-specific, shorter subnet mask. CIDR is a form of route summarization and is synonymous with the term supernetting.

CIDR ignores the limitation of classful boundaries, and allows summarization with masks that are smaller than that of the default classful mask. This type of summarization helps reduce the number of entries in routing updates and lowers the number of entries in local routing tables. It also helps reduce bandwidth utilization for routing updates and results in faster routing table lookups.

In the figure below, R1 requires a summary static route to reach networks in the range of 172.20.0.0/16 to 172.23.0.0/16.



Summarizing networks into a single address and mask can be done in three steps:

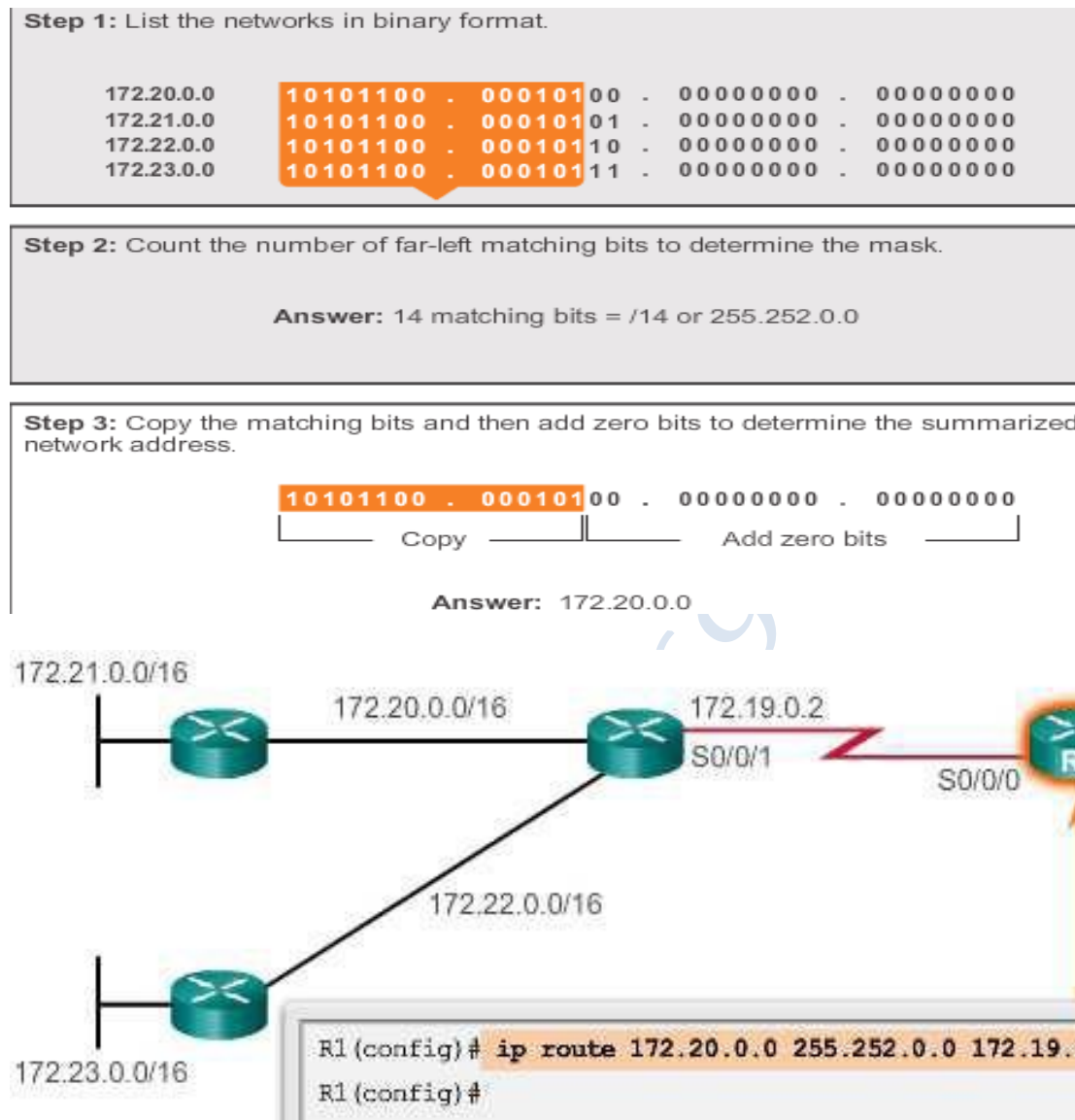
Step 1. List the networks in binary format. Figure below lists networks 172.20.0.0/16 to 172.23.0.0/16 in binary format.

Step 2. Count the number of far-left matching bits to determine the mask for the summary route. Figure below highlights the 14 far left matching bits. This is the prefix, or subnet mask, for the summarized route: /14 or 255.252.0.0.

Step 3. Copy the matching bits and then add zero bits to determine the summarized network address. Figure below shows that the matching bits with zeros at the end results in the network address 172.20.0.0. The four networks - 172.20.0.0/16, 172.21.0.0/16, 172.22.0.0/16, and 172.23.0.0/16 - can be

summarized into the single network address and prefix 172.20.0.0/14.

Figure below displays R1 configured with a summary static route to reach networks 172.20.0.0/16 to 172.23.0.0/16.



62. TOPIC

Synopsis/ Description

- Route Summarization
- Configuration of IPv4 Route Summarization
- Verifying Connectivity

Dynamic routing protocols have been used in networks since the late 1980s. One of the first routing protocols was Routing Information Protocol (RIP). RIP version 1 (RIPv1) was released in 1988. As networks evolved and became more complex, new routing protocols emerged. The RIP routing protocol was updated to accommodate growth in the network environment, into RIPv2. However, the newer version of RIP still does not scale to the larger network implementations of today. To address the needs

of larger networks, two advanced routing protocols were developed: Open Shortest Path First (OSPF) and Intermediate System-to-Intermediate System (IS-IS). Cisco developed the Interior Gateway Routing Protocol (IGRP) and Enhanced IGRP (EIGRP), which also scales well in larger network implementations.

Additionally, there was the need to connect different internetworks and provide routing between them. The Border Gateway Protocol (BGP) is now used between Internet service providers (ISPs). BGP is also used between ISPs and their larger private clients to exchange routing information.

	Interior Gateway Protocols				Exterior Gateway Protocols
	Distance Vector		Link-State		Path Vector
IPv4	RIPv2	EIGRP	OSPFv2	IS-IS	BGP-4
IPv6	RIPng	EIGRP for IPv6	OSPFv3	IS-IS for IPv6	BGP-MP

Routing protocols are used to facilitate the exchange of routing information between routers. A routing protocol is a set of processes, algorithms, and messages that are used to exchange routing information and populate the routing table with the routing protocols choice of best paths. The purpose of dynamic routing protocols includes:

- Discovery of remote networks
 - Maintaining up-to-date routing information
 - Choosing the best path to destination networks
 - Ability to find a new best path if the current path is no longer available
- The main components of dynamic routing protocols include:

Data structures - Routing protocols typically use tables or databases for its operations. This information is kept in RAM.

Routing protocol messages - Routing protocols use various types of messages to discover neighboring routers, exchange routing information, and other tasks to learn and maintain accurate information about the network.

Algorithm - An algorithm is a finite list of steps used to accomplish a task. Routing protocols use algorithms for facilitating routing information and for best path determination.

63. TOPIC

Synopsis/Description

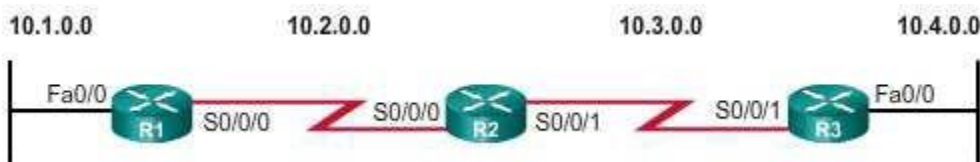
- I. Protocol Fundamentals
- II. Routing Protocols

All routing protocols are designed to learn about remote networks and to quickly adapt whenever there is a change in the topology. The method that a routing protocol uses to accomplish this depends upon the algorithm it uses and the operational characteristics of that protocol.

In general, the operations of a dynamic routing protocol can be described as follows:

1. The router sends and receives routing messages on its interfaces.
2. The router shares routing messages and routing information with other routers that are using the same routing protocol.
3. Routers exchange routing information to learn about remote networks.
4. When a router detects a topology change the routing protocol can advertise this change to other routers.

All routing protocols follow the same patterns of operation. To help illustrate this, consider the following scenario in which all three routers are running RIPv2.



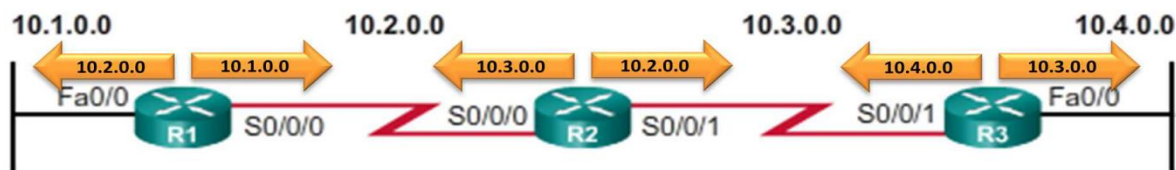
When a router powers up, it knows nothing about the network topology. It does not even know that there are devices on the other end of its links. The only information that a router has is from its own saved configuration file stored in NVRAM. After a router boots successfully, it applies the saved configuration. If the IP addressing is configured correctly, then the router initially discovers its own directly connected networks.

Routers proceed through the boot up process and then discovers any directly connected networks and subnet masks. This information is added to their routing tables as follows:

-
- R1 adds the 10.1.0.0 network available through interface FastEthernet 0/0 and 10.2.0.0 is available through interface Serial 0/0/0.
- R2 adds the 10.2.0.0 network available through interface Serial 0/0/0 and 10.3.0.0 is available through interface Serial 0/0/1.
- R3 adds the 10.3.0.0 network available through interface Serial 0/0/1 and 10.4.0.0 is available through interface FastEthernet 0/0.

With this initial information, the routers then proceed to find additional route sources for their routing tables.

After initial boot up and discovery, the routing table is updated with all directly connected networks and the interfaces those networks reside on.



	Network	Interface	Hop
C	10.1.0.0	Fa0/0	0
C	10.2.0.0	S0/0/0	0

	Network	Interface	Hop
C	10.2.0.0	S0/0/0	0
C	10.3.0.0	S0/0/1	0

	Network	Interface	Hop
C	10.3.0.0	S0/0/1	0
C	10.4.0.0	Fa0/0	0

Distance vector routing protocols implement a routing loop prevention technique known as **split horizon**. It prevents information from being sent out the same interface from which it was received. For example, R2 does not send an update containing the network 10.1.0.0 out of Serial 0/0/0, because R2 learned about network 10.1.0.0 through Serial 0/0/0.

If a routing protocol is configured, the next step is for the router to begin exchanging routing updates to learn about any remote routes.

The router sends an update packet out all interfaces that are enabled on the router. The update contains the information in the routing table, which currently are all directly connected networks.

At the same time, the router also receives and processes similar updates from other connected routers. Upon receiving an update, the router checks it for new network information. Any networks that are not currently listed in the routing table are added.

Refer to the figure above for a topology setup between three routers, R1, R2, and R3. Based on this topology, below is a listing of the different updates that R1, R2, and R3 send and receive during initial convergence.

R1:

R2:

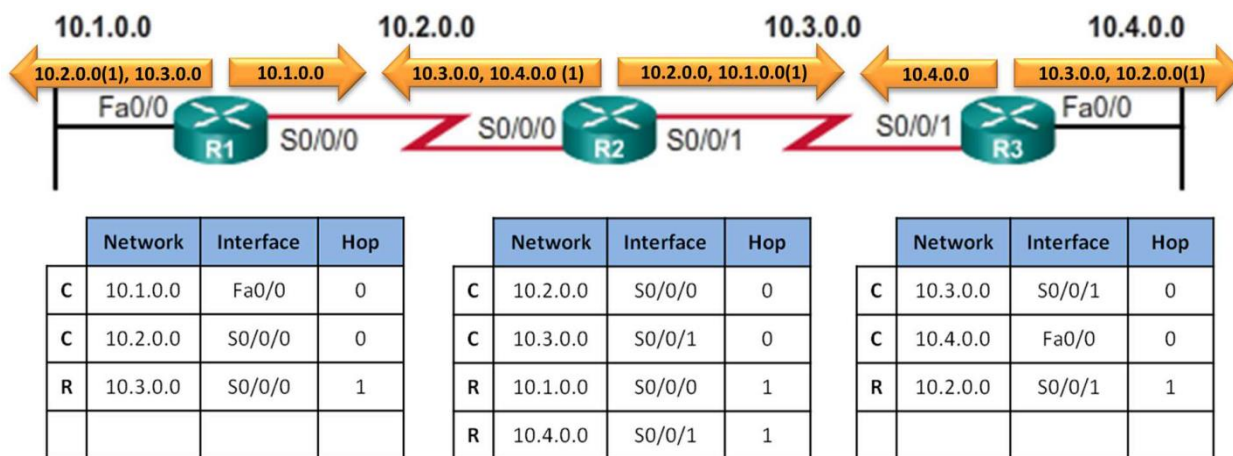
R3:

- Sends an update about network 10.1.0.0 out the Serial0/0/0 interface
- Sends an update about network 10.2.0.0 out the FastEthernet0/0 interface
- Receives update from R2 about network 10.3.0.0 and increments the hop count by 1
- Stores network 10.3.0.0 in the routing table with a metric of 1
- Sends an update about network 10.3.0.0 out the Serial 0/0/0 interface
- Sends an update about network 10.2.0.0 out the Serial 0/0/1 interface
- Receives an update from R1 about network 10.1.0.0 and increments the hop count by 1
- Stores network 10.1.0.0 in the routing table with a metric of 1
- Receives an update from R3 about network 10.4.0.0 and increments the hop count by 1
- Stores network 10.4.0.0 in the routing table with a metric of 1
- Sends an update about network 10.4.0.0 out the Serial 0/0/1 interface
- Sends an update about network 10.3.0.0 out the FastEthernet0/0

- Receives an update from R2 about network 10.2.0.0 and increments the hop count by 1
- Stores network 10.2.0.0 in the routing table with a metric of 1

After this first round of update exchanges, each router knows about the connected networks of their directly connected neighbors. However, did you notice that R1 does not yet know about 10.4.0.0 and that R3 does not yet know about 10.1.0.0? Full knowledge and a converged network do not take place until there is another exchange of routing information.

At this point the routers have knowledge about their own directly connected networks and about the connected networks of their immediate neighbors. Continuing the journey toward convergence, the routers exchange the next round of periodic updates. Each router again checks the updates for new information.



Refer to the figure above for a topology setup between three routers, R1, R2, and R3. After initial discovery is complete, each router continues the convergence process by sending and receiving the following updates.

R1:

R2:

- Sends an update about network 10.1.0.0 out the Serial 0/0/0 interface
- Sends an update about networks 10.2.0.0 and 10.3.0.0 out the FastEthernet0/0 interface
- Receives an update from R2 about network 10.4.0.0 and increments the hop count by 1
- Stores network 10.4.0.0 in the routing table with a metric of 2
- Same update from R2 contains information about network 10.3.0.0 with a metric of 1. There is no change; therefore, the routing information remains the same
- Sends an update about networks 10.3.0.0 and 10.4.0.0 out of Serial 0/0/0 interface
- Sends an update about networks 10.1.0.0 and 10.2.0.0 out of Serial 0/0/1 interface
- Receives an update from R1 about network 10.1.0.0. There is no change; therefore, the routing information remains the same

R3:

- Receives an update from R3 about network 10.4.0.0. There is no change; therefore, the routing information remains the same

- Sends an update about network 10.4.0.0 out the Serial 0/0/1 interface
- Sends an update about networks 10.2.0.0 and 10.3.0.0 out the FastEthernet0/0 interface
- Receives an update from R2 about network 10.1.0.0 and increments the hop count by 1
- Stores network 10.1.0.0 in the routing table with a metric of 2
- Same update from R2 contains information about network 10.2.0.0 with a metric of 1. There is no change; therefore, the routing information remains the same

Distance vector routing protocols typically implement a routing loop prevention technique known as split horizon. Split horizon prevents information from being sent out the same interface from which it was received. For example, R2 does not send an update containing the network 10.1.0.0 out of Serial 0/0/0, because R2 learned about network 10.1.0.0 through Serial 0/0/0.

After routers within a network have converged, the router can then use the information within the route table to determine the best path to reach a destination. Different routing protocols have different ways of calculating the best path.

64. TOPIC

Synopsis/Description

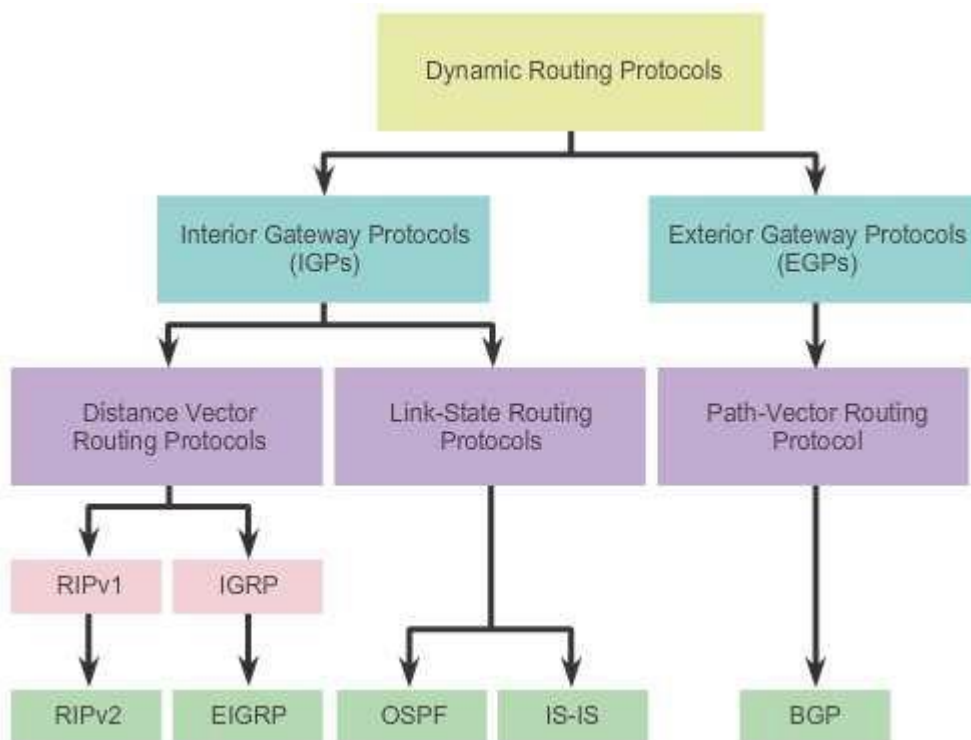
- I. Different Routing Protocols
- II. Classful Routing protocols
- III. Interior Gateway Protocols
- IV. Exterior Gateway Protocols

Routing protocols can be classified into different groups according to their characteristics. Specifically, routing protocols can be classified by their:

- **Purpose** - Interior Gateway Protocol (IGP) or Exterior Gateway Protocol (EGP)
- **Operation** - Distance vector, link-state protocol, or path-vector protocol
- **Behavior** - Classful (legacy) or classless protocol
- For example, IPv4 routing protocols are classified as follows:
- **RIPv1 (legacy)** - IGP, distance vector, classful protocol
- **IGRP (legacy)** - IGP, distance vector, classful protocol developed by Cisco (deprecated from 12.2 IOS and later)
- **RIPv2** - IGP, distance vector, classless protocol
- **EIGRP** - IGP, distance vector, classless protocol developed by Cisco
- **OSPF** - IGP, link-state, classless protocol
- **IS-IS** - IGP, link-state, classless protocol
- **BGP** - EGP, path-vector, classless protocol

The classful routing protocols, RIPv1 and IGRP, are legacy protocols and are only used in older networks. These routing protocols have evolved into the classless routing protocols, RIPv2 and EIGRP, respectively. Link-state routing protocols are classless by nature.

Figure below displays a hierarchical view of dynamic routing protocol classification.



An autonomous system (AS) is a collection of routers under a common administration such as a company or an organization. An AS is also known as a routing domain. Typical examples of an AS are a company's internal network and an ISP's network.

The Internet is based on the AS concept; therefore, two types of routing protocols are required:

Interior Gateway Protocols (IGP) - Used for routing within an AS. It is also referred to as intra-AS routing. Companies, organizations, and even service providers use an IGP on their internal networks. IGPs include RIP, EIGRP, OSPF, and IS-IS.

Exterior Gateway Protocols (EGP) - Used for routing between AS. It is also referred to as inter-AS routing. Service providers and large companies may interconnect using an EGP. The Border Gateway Protocol (BGP) is the only currently-viable EGP and is the official routing protocol used by the Internet.

Note: Because BGP is the only EGP available, the term EGP is rarely used; instead, most engineers simply refer to BGP.

Distance Vector Routing Protocols

Distance vector means that routes are advertised by providing two characteristics:

Distance - Identifies how far it is to the destination network and is based on a metric such as the hop count, cost, bandwidth, delay, and more.

Vector - Specifies the direction of the next-hop router or exit interface to reach the destination.

A router using a distance vector routing protocol does not have the knowledge of the entire path to a destination network. Distance vector protocols use routers as sign posts along the path to the final destination. The only

information a router knows about a remote network is the distance or metric to reach that network and which path or interface to use to get there. Distance vector routing protocols do not have an actual map of the network topology.

There are four distance vector IPv4 IGPs:

- **RIPv1** - First generation legacy protocol
- **RIPv2** - Simple distance vector routing protocol
- **IGRP** - First generation Cisco proprietary protocol (obsolete and replaced by EIGRP)
- **EIGRP** - Advanced version of distance vector routing

Link State Routing Protocols

In contrast to distance vector routing protocol operation, a router configured with a link-state routing protocol can create a complete view or topology of the network by gathering information from all of the other routers.

RIP-enabled routers send periodic updates of their routing information to their neighbors. Link-state routing protocols do not use periodic updates. After the network has converged, a link-state update is only sent when there is a change in the topology.

Link-state protocols work best in situations where:

- The network design is hierarchical, usually occurring in large networks
 - Fast convergence of the network is crucial
 - The administrators have good knowledge of the implemented link-state routing protocol
- There are two link-state IPv4 IGPs:
- **OSPF** - Popular standards based routing protocol
 - **IS-IS** - Popular in provider networks

65. TOPIC

Synopsis/Description

- I. Network Convergence
- II. Routing Protocols and Network Convergence

Convergence is the process of routers agreeing on optimal routes for forwarding packets and thereby completing the updating of their routing tables. When this occurs, each router independently runs a routing algorithm to recalculate metrics and build a new routing table based on this information.

The network has converged when all routers have complete and accurate information about the entire network. Convergence time is the time it takes routers to share information, calculate best paths, and update their routing tables. A network is not completely operable until the network has converged; therefore, most networks require short convergence times.

Convergence is both collaborative and independent. The routers share information with each other, but must independently calculate the impacts of the topology change on their own routes. Because they develop an agreement with the new topology independently, they are said to converge on this consensus.

Convergence properties include the speed of propagation of routing information and the calculation of optimal paths. The speed of propagation refers to the amount of time it takes for routers within the network to forward routing information.

Routing protocols can be rated based on the speed to convergence; the faster the convergence, the better the routing protocol. Generally, older protocols, such as RIP, are slow to converge, whereas modern protocols, such as EIGRP and OSPF, converge more quickly.

66. TOPIC

Distance vector routing protocols share updates between neighbors. Neighbors are routers that share a link and are configured to use the same routing protocol. The router is only aware of the network addresses of its own interfaces and the remote network addresses it can reach through its neighbors. Routers using distance vector routing are not aware of the network topology.

Some distance vector routing protocols send periodic updates. For example, RIP sends a periodic update to all of its neighbors every 30 seconds. RIP does this even if the topology has not changed; it continues to send updates. RIPv1 reaches all of its neighbors by sending updates to the all-hosts IPv4 address of 255.255.255.255, a broadcast.

The broadcasting of periodic updates is inefficient because the updates consume bandwidth and consume network device CPU resources. Every network device has to process a broadcast message. RIPv2 and EIGRP, instead, use multicast addresses so that only neighbors that need updates will receive them. EIGRP can also send a unicast message to only the affected neighbor. Additionally, EIGRP only sends an update when needed, instead of periodically.

The two modern IPv4 distance vector routing protocols are RIPv2 and EIGRP. RIPv1 and IGRP are listed only for historical accuracy.

IGRP

The Interior Gateway Routing Protocol (IGRP) was the first proprietary IPv4 routing protocol developed by Cisco in 1984. It used the following design characteristics:

Bandwidth, delay, load, and reliability are used to create a composite metric.

Routing updates are broadcast every 90 seconds, by default.

In 1992, IGRP was replaced by Enhanced IGRP (EIGRP). Like RIPv2, EIGRP also introduced support for VLSM and CIDR. EIGRP increases efficiency, reduces routing updates, and supports secure message exchange.

EIGRP also introduced:

Bounded triggered updates - It does not send periodic updates. Only routing table changes are propagated, whenever a change occurs. This reduces the amount of load the routing protocol places on the network. Bounded triggered updates means that EIGRP only sends to the neighbors that need it. It uses less bandwidth, especially in large networks with many routes.

Hello keepalive mechanism - A small Hello message is periodically exchanged to maintain adjacencies with neighboring routers. This means a very low usage of network resources during normal operation, instead of the periodic updates.

Maintains a topology table - Maintains all the routes received from neighbors (not only the best paths) in a topology table. DUAL can insert backup routes into the EIGRP topology table.

Rapid convergence - In most cases, it is the fastest IGP to converge because it maintains alternate routes, enabling almost instantaneous convergence. If a primary route fails, the router can use the alternate route identified. The switchover to the alternate route is immediate and does not involve interaction with other routers.

CS407

[Routing and switching]

[Reading material]

Made by Mah jabeen

[mahjabeen97869@gmail.com]